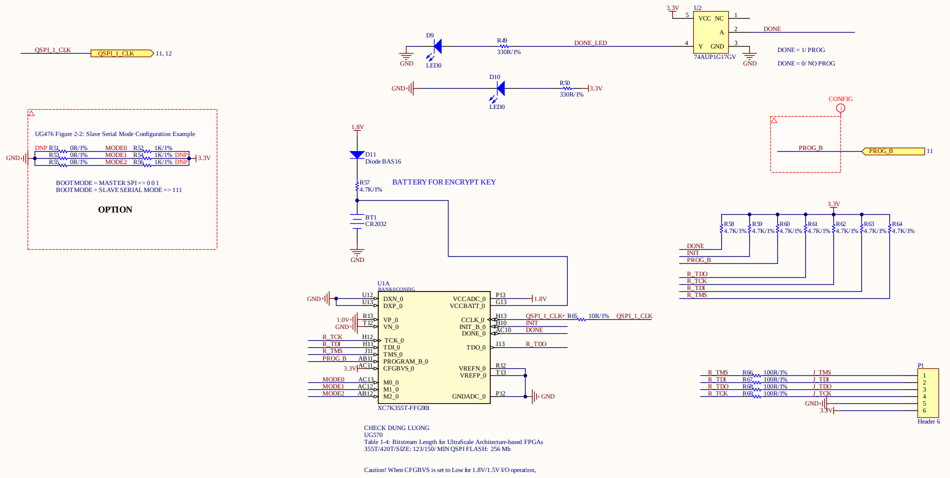

project: ***HTPSR5G***

Size: A4 | Number: 10 | Revision: V7

Date: 26/01/2021 Time: 3:09:08 CH | Sheet 10 of 51

File: P10 CONFIG.SchDoc

HTP

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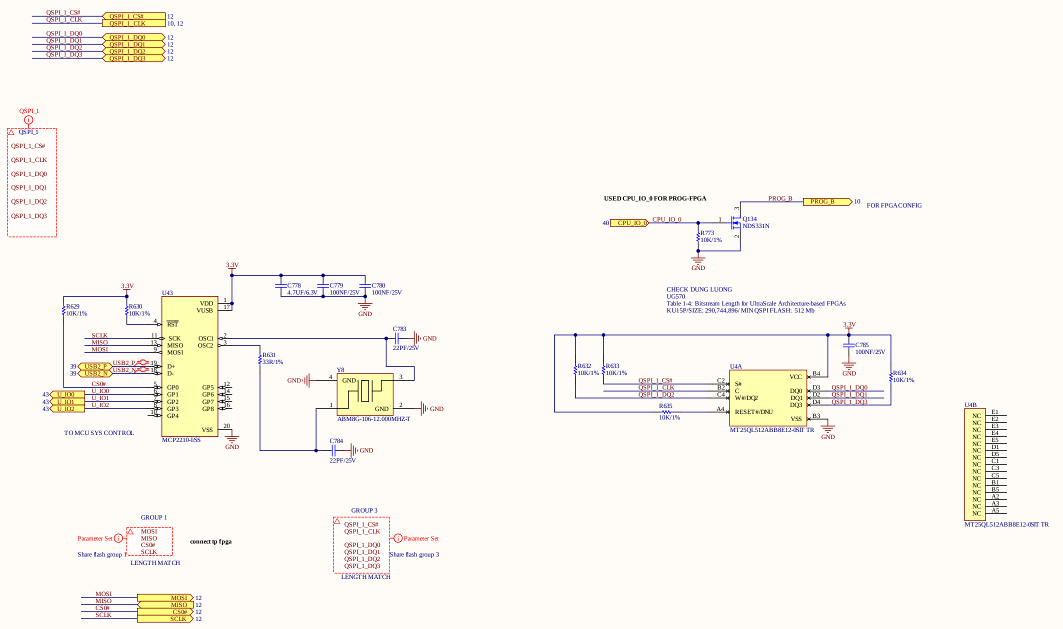
Hanoi



project: **HTPSR5G**

Size: A4	Number: 11	Revision: V7
Date: 26/01/2021	Time: 3:09:08 CH	Sheet 11 of 51
File: P11 QSPI FLASH.SchDoc		

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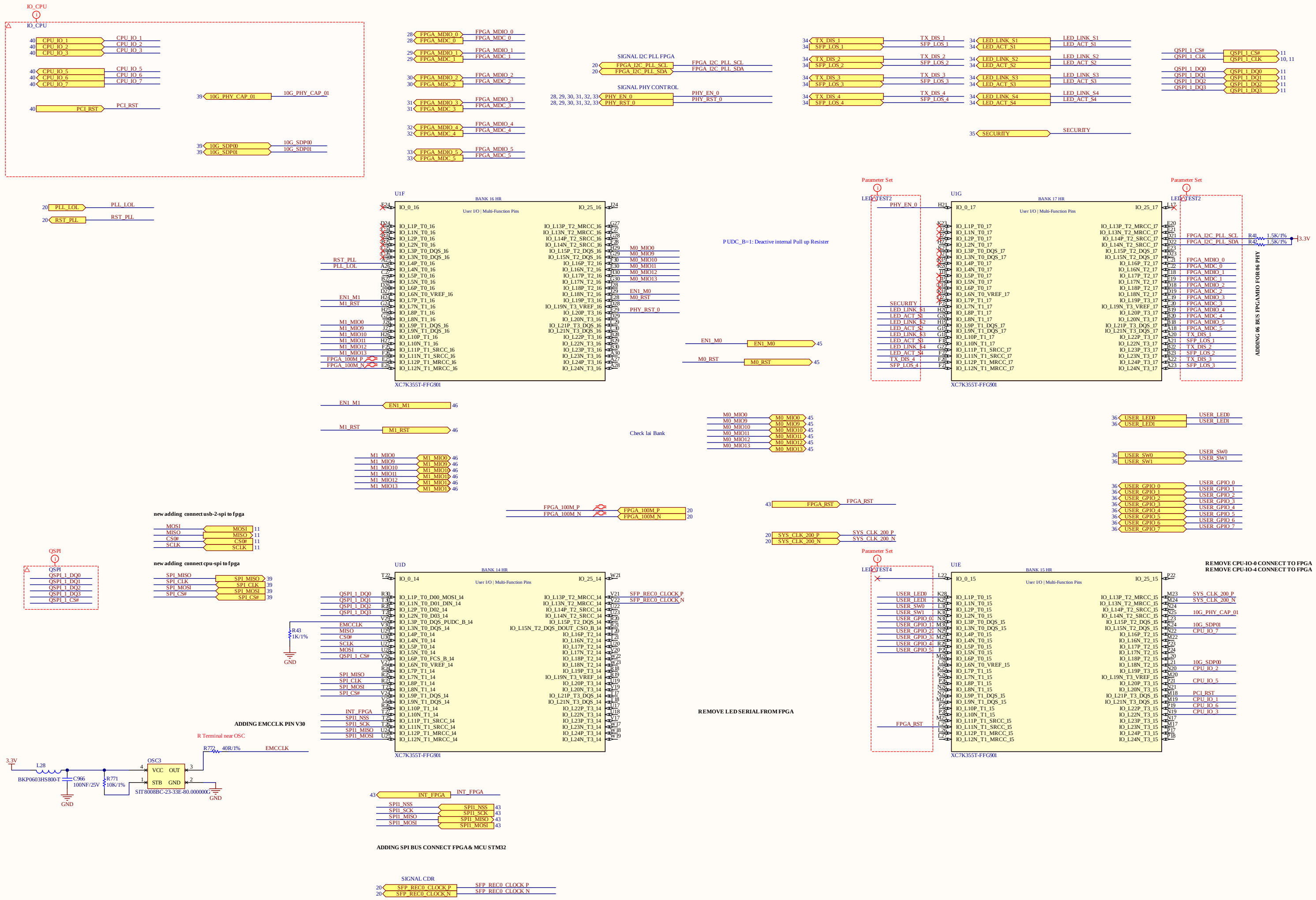
Hanoi

HTP



project: **HTPSR5G**

Size: A4	Number: 12	Revision: V7
Date: 26/01/2021	Time: 3:09:09 CH	Sheet 12 of 51
File: P12 BANKS HR 14 15 16 17.SchDoc		

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BANK 16 & 17 1.8V FOR QDR II

D and BW0/1 in same bank:0k
Only one write data byte is allowed per byte lane: 0k
All byte lanes that are used for the write data of a single component must be adjacent, no skip byte lanes are allowed: ok

One of the write data bytes of a memory component should be allocated in the: ok
Memory Clock pair must be allocated in one of the byte lanes that are used for the write data of the corresponding memory component.: ok

Memory clock should come from one of the center byte lanes (byte lanes 1 and 2).: ok
K/K# can be allocated to any PN pair: ok

All address/control (A/C) bits must be allocated in a single bank: ok
All A/C byte lanes should be contiguous and no skip byte lanes is allowed: ok
The address/control bank should be the same or adjacent to that of the write data bank: ok
There should not be any empty byte lane or read byte lane between A/C and write data byte lanes. This rule applies when A/C and write data share the same bank or allocated in adjacent banks: ok
Address/control pins should not share a byte lane with the write data as well as rea: ok

The entire read data bus must be placed in a single bank irrespective of the number of memory components.: ok

All byte lanes that are used for the read data of a single component must be adjacent, no skip byte lanes are allowed.: ok

One of the read data bytes of a memory component should be allocated in the center byte lanes (byte lanes 1 and 2).: ok

If a byte lane is used for read data, Bit[0] and Bit[6] must be used. Read clock (CQ or CQ#) gets the first priority and data (Q) is the next.: ok

Read data buses of two components should not share a byte lane.: ok

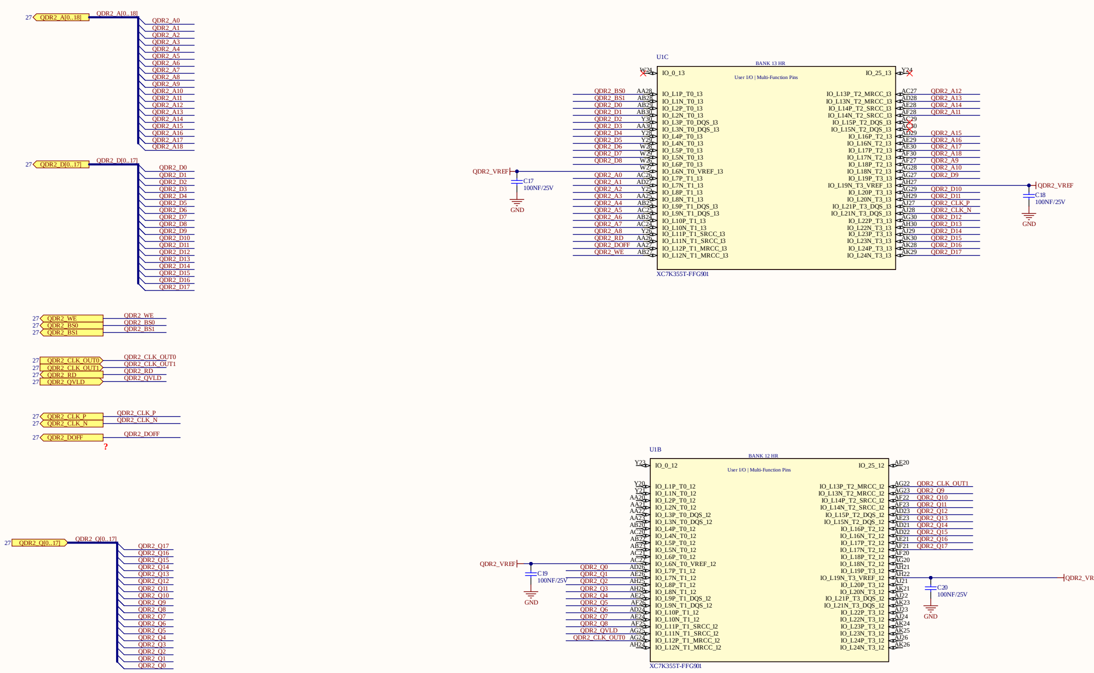
Read Clock pair must be allocated in one of the byte lanes that are used for the read: ok

CQ/CQ# pair must be allocated in a single byte lane.: ok

CQ/CQ# must be allocated only in the byte lanes 1 and 2 because the other byte: ok

CQ/CQ# must be allocated in the center byte lane of all used byte lanes. If two byte lanes are used for the read data, either one of them can be used for CQ/CQ# allocation: ok

CQ and CQ# must be allocated to either pin 0 or pin 6 of a byte lane. For example, if CQ is allocated to pin 0, CQ# should be allocated to pin 6 and vice versa: ok



Pins can swap freely within each Write Data byte group.
• Pins can swap freely within each Read Data byte group, except CQ/CQ# pins. Pins can swap freely within and between their corresponding byte groups, but should not violate above mentioned Read Data pin/byte lane allocation rules.
• Pins can swap freely withing each Address/Control byte group. Pins can swap freely within and between their corresponding byte groups, but should not violate above mentioned Address/Control pin/byte lane allocation rules.
• Write Data Byte groups can swap easily with each other, but should not violate above mentioned Write Data pin/byte lane allocation rules.
• Read Data Byte groups can swap easily with each other, but should not violate above mentioned Read Data pin/byte lane allocation rules.
• Address/Control Byte groups can swap easily with each other, but should not violate above mentioned Address/Control pin/byte lane allocation rules.
• No other pin swapping is permitted.

HTP

project: HTPSR5G

Size: A4 | Number: 13 | Revision: V7

Date: 26/01/2021 Time: 3:09:10 CH | Sheet 13 of 51

File: P13 BANKS HR 12 13.SchDoc

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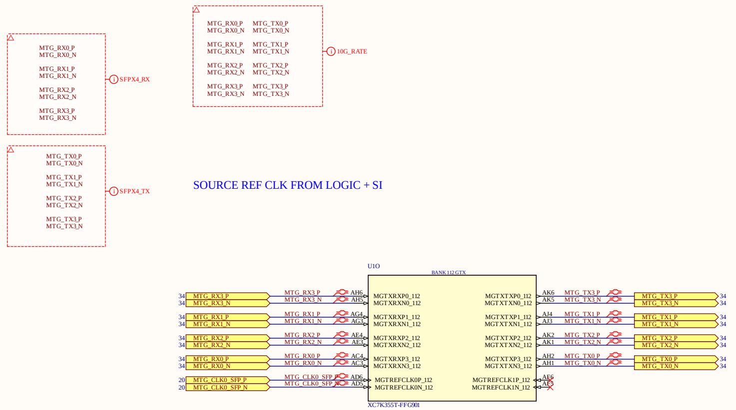
Hanoi



HTP

project: *HTPSR5G*

Size: A4 | Number: 14 | Revision: V7

Date: 26/01/2021 Time: 3:09:10 CH Sheet 14 of 51

File: P14 GTX BANKS SFP&SFP+.SchDoc

HTP Hitech Co.,Ltd
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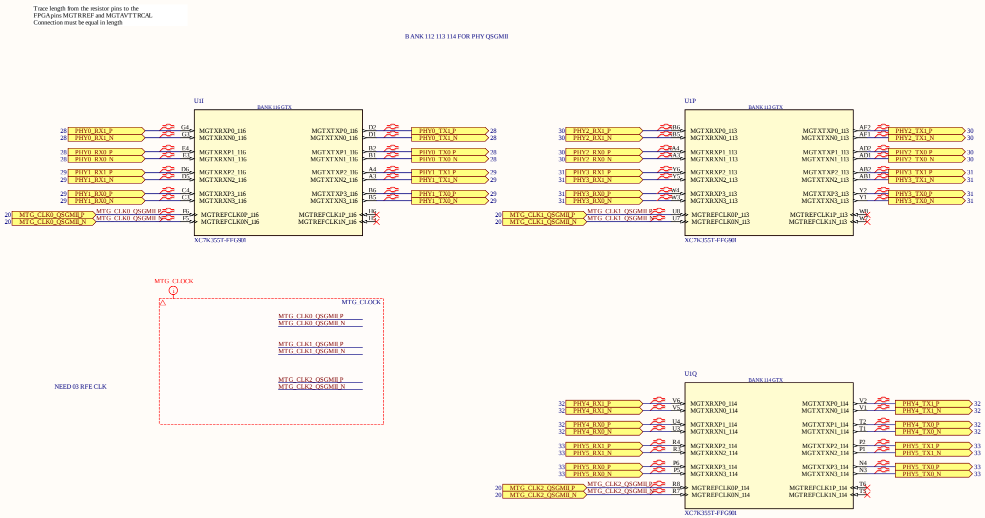
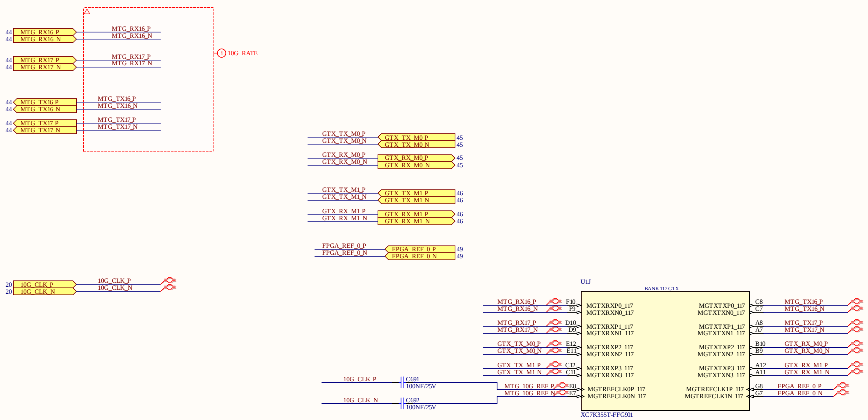


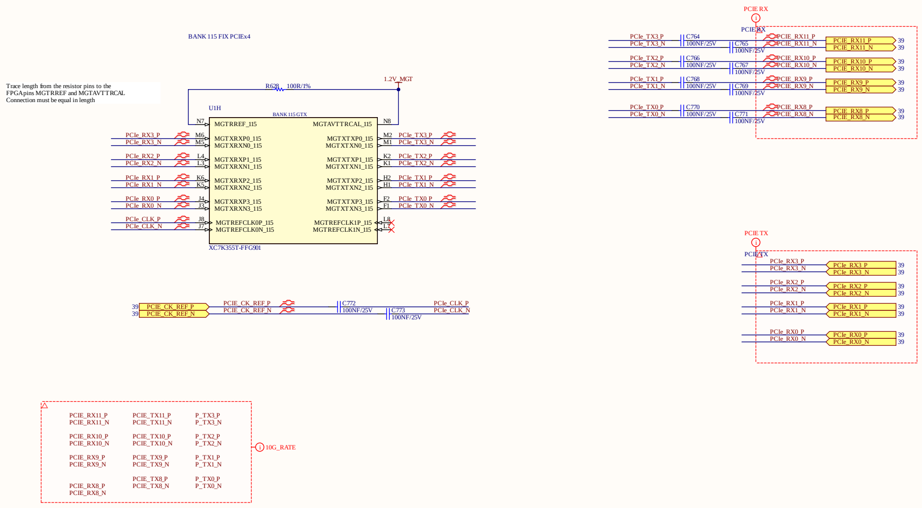
project: *HTPSR5G*			**HTP Hitech Co.,Ltd**
Size: A4	Number: 15	Revision: V7	OCT5B - 2303, Resco Urban Area
Date: 26/01/2021	Time: 3:09:10 CH	Sheet 15 of 51	74 Pham Van Dong, Bac Tu Liem
File: P15 GTX BANKS QSGMII.SchDoc			Hanoi

HTP



project: **HTPSR5G**			**HTP Hitech Co.,Ltd**
Size: A4	Number: 16	Revision: V7	*OCT5B - 2303, Resco Urban Area*
Date: 26/01/2021	Time: 3:09:11 CH	Sheet 16 of 51	*74 Pham Van Dong, Bac Tu Liem*
File: P16 GTX BANK 10G KR.SchDoc			*Hanoi*

HTP



HTP

project: *HTPSR5G*

Size: A4	Number: 17	Revision: V7
Date: 26/01/2021	Time: 3:09:11 CH	Sheet 17 of 51

File: P17 GTX BANK PCIe x4.SchDoc

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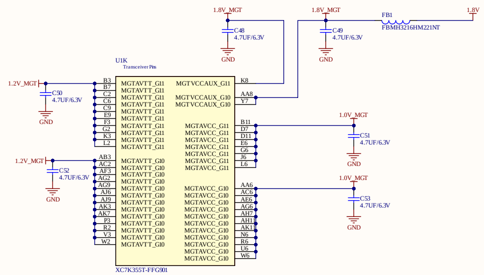
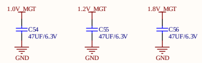
Hanoi

GTH Transceiver PCB Capacitor Recommendations
UG476 TABLE 5-7
USED 03 CAP 4.7UF for total



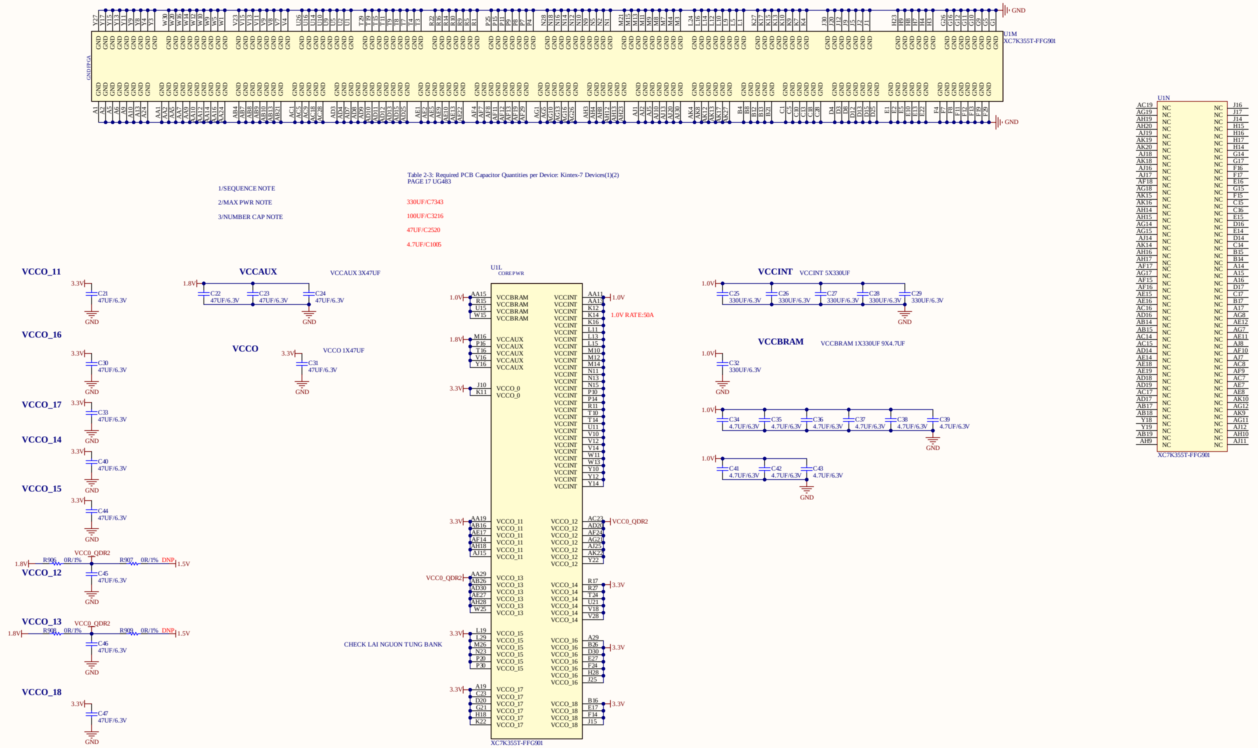
SHARE BULL CAPS



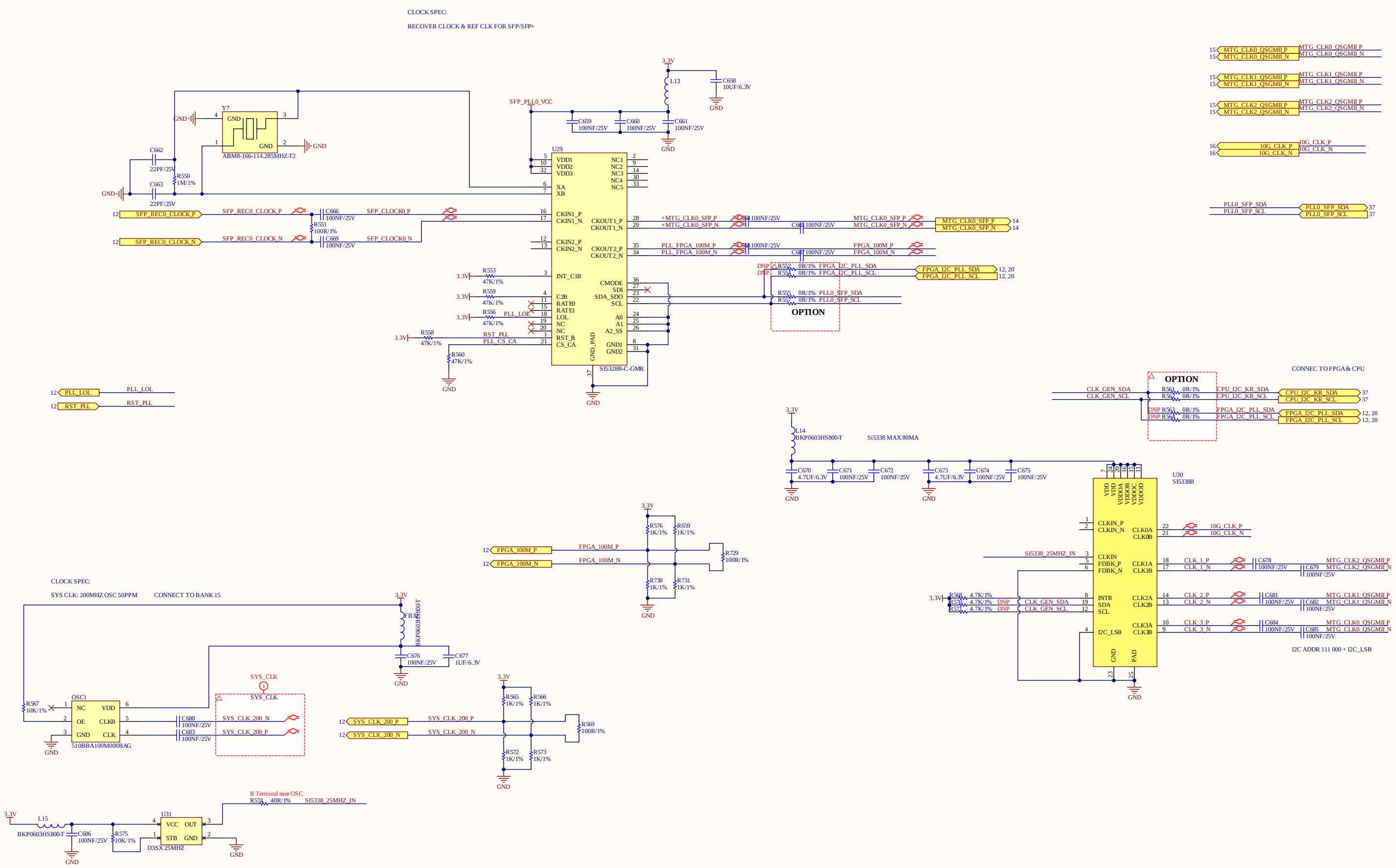
GROUP10	GROUP11
CAPSBANK112	CAPSBANK115
CAPSBANK113	CAPSBANK116
CAPSBANK114	CAPSBANK117

project: **HTPSR5G**			**HTP Hitech Co.,Ltd** *OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi*
Size: A4	Number: 18	Revision: V7	
Date:26/01/2021 Time: 3:09:11 CH		Sheet 18of 51	
File: P18 GTX PWR.SchDoc			

HTP

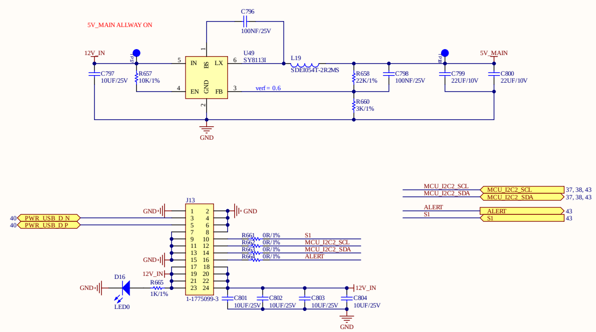


HTP	project: HTPSR5G		HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
	Size: A4	Number: 19	Revision: V7
	Date:26/01/2021 Time: 3:09:11 CH		Sheet 19of 51
	File: P19 FPGA PWR.SchDoc		



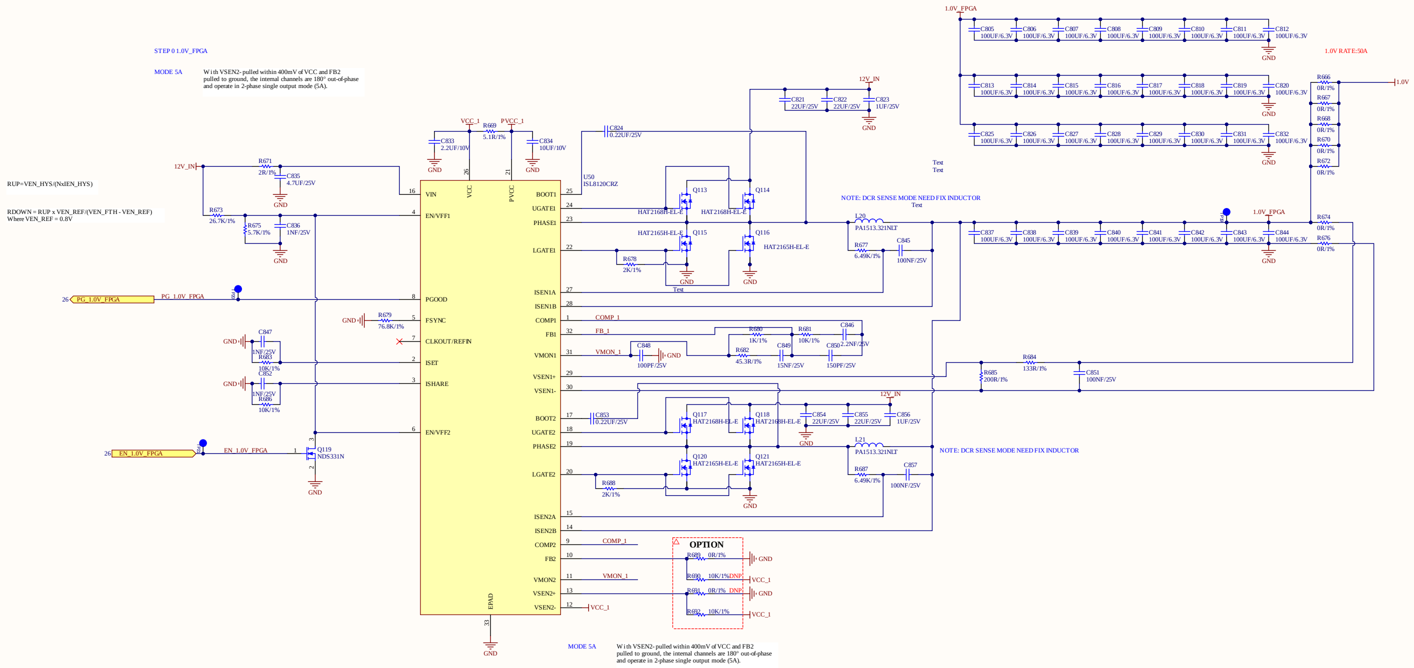
project: HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
Size: A4	Number: 20	Revision: V7	
Date: 26/01/2021	Time: 3:09:12 CH	Sheet 20of 51	
File: P20 CLOCK SYS.SchDoc			

HTP



project: **HTPSR5G**			HTP Hitech Co.,Ltd
Size: A4	Number: 21	Revision: V7	OCT5B - 2303, Resco Urban Area
Date: 26/01/2021	Time: 3:09:12 CH	Sheet 21 of 51	74 Pham Van Dong, Bac Tu Liem
File: P21 SYSTEM PWR.SchDoc			Hanoi

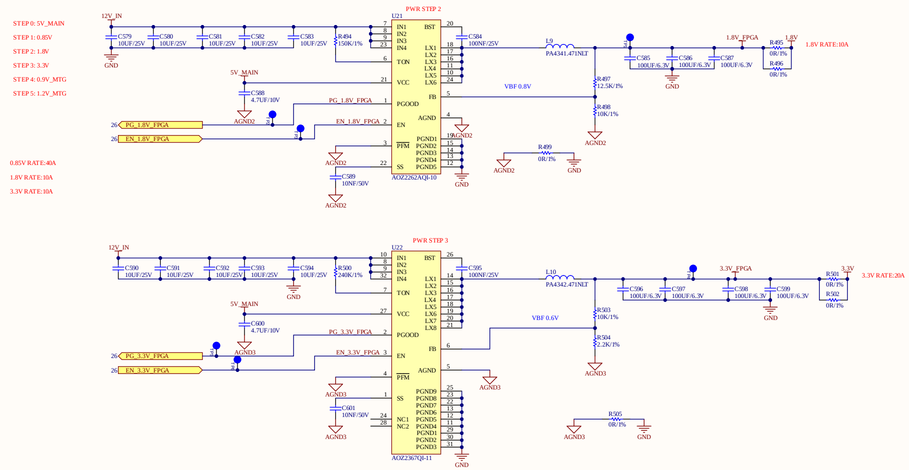
HTP



Test

project: *HTPSR5G*			**HTP Hitech Co.,Ltd**
Size: A4	Number: 22	Revision: V7	*OCT5B - 2303, Resco Urban Area*
Date:26/01/2021 Time: 3:09:12 CH		Sheet 22of 51	*74 Pham Van Dong, Bac Tu Liem*
File: P22 PWR CORE 50A.SchDoc			*Hanoi*

HTP



project: **HTPSR5G**		HTP Hitech Co.,Ltd
Size: A4	Number: 23	Revision: V7
Date: 26/01/2021	Time: 3:09:13 CH	Sheet 23 of 51
File: P23 PWR FPGA 1.8V 3.3V.SchDoc		

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Hanoi

HTP

PWR SEQUENCE GTX:

1/ VCCINT: 1.0V

2/ VMGTAVCC: 1V

3/ VMGTAVTT: 1.2V & VMGTVCCAUX: 1.8V

PWR RATE GTX:

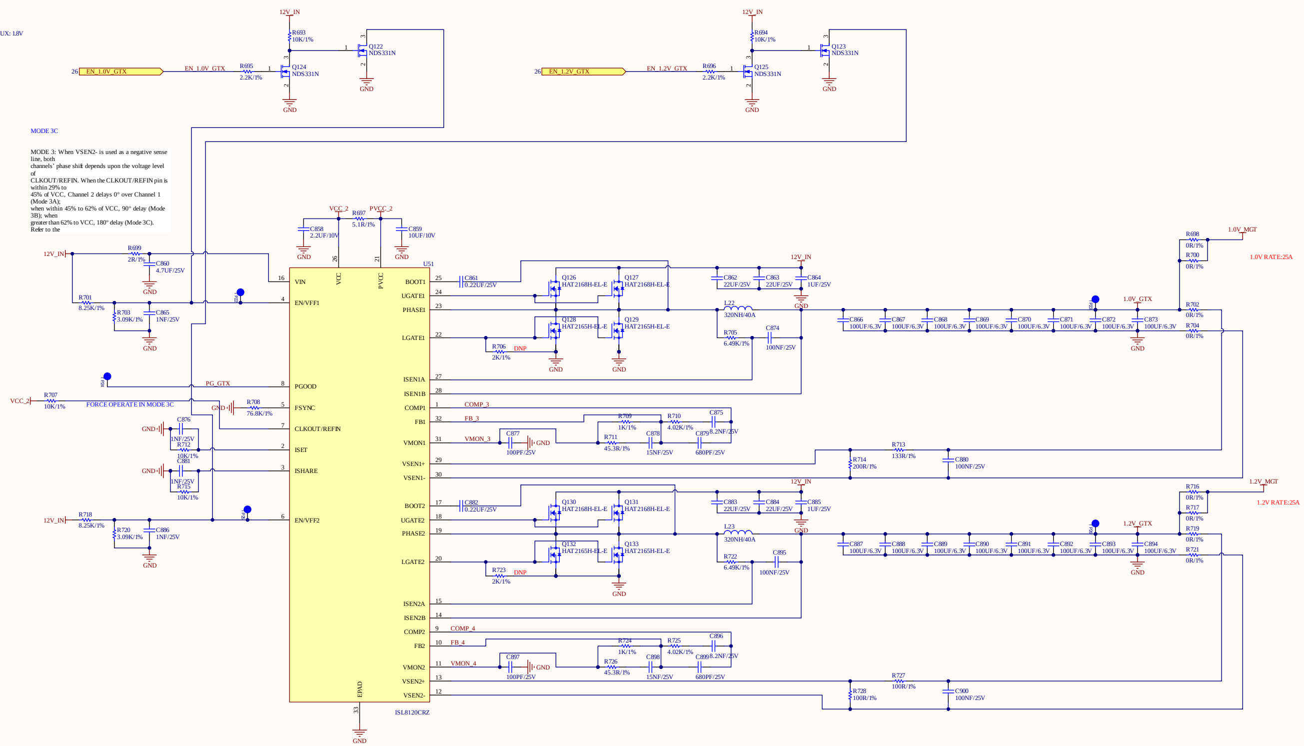
1/ VMGTAVCC: 1V/8A

2/ MGTVCCAUX: 1.8V SHARE

3/ VMGTAVTT: 1.2V/8A

MODE 3C

MODE 3: When VSEN2- is used as a negative sense line, both channels' phase shift depends upon the voltage level of CLKOUT/REFIN. When the CLKOUT/REFIN pin is within 29% to 45% of VCC, Channel 2 delays 0° over Channel 1 (Mode 3A); when within 45% to 62% of VCC, 90° delay (Mode 3B); when greater than 62% to VCC, 180° delay (Mode 3C). Refer to the



HTP	project: HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
	Size: A4	Number: 24	Revision: V7	
	Date: 26/01/2021	Time: 3:09:13 CH	Sheet 24 of 51	
	File: P24 PWR GTX 25A.SchDoc			

PWR SEQUENCE PHY:

1/ 1.05V_PHY

2/ 3.3V_PHY

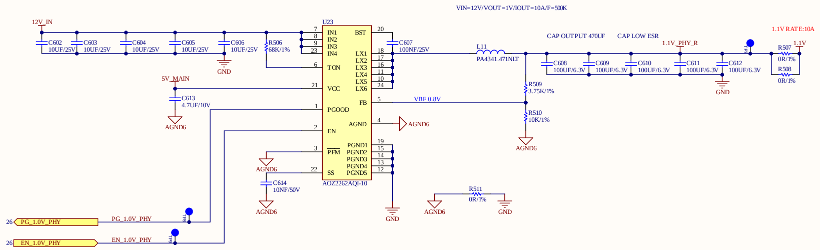
PWR RATE FPGA:

3/ 3.3V_PHY SHARE 3.3V

2/ 1.05V_PHY/5A   PHY 1.05V RATE 5A, 8X0.6=4.8A

3.3V_PHY SHARE 3.3V SYSTEM

1.0V_PHY RATE 5A

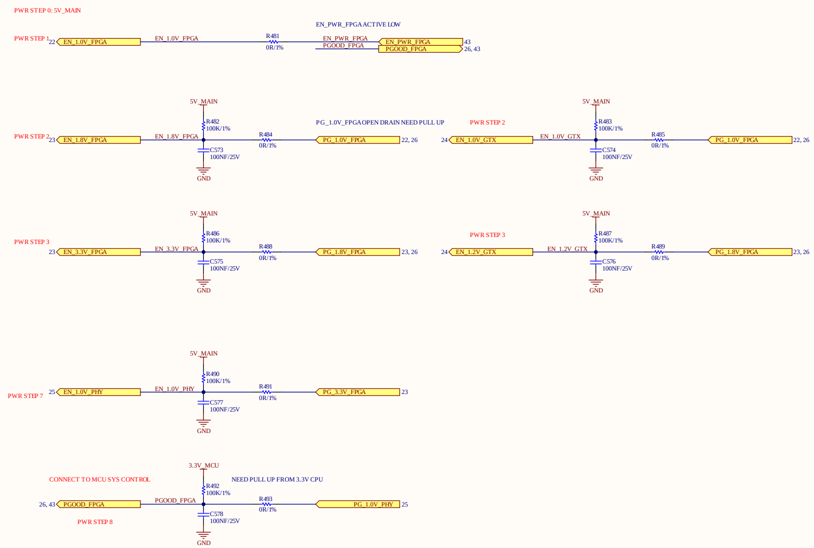


project: HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
Size: A4	Number: 25	Revision: V7	
Date: 26/01/2021 Time: 3:09:13 CH		Sheet 25 of 51	
File: P25 PWR PHY.SchDoc			

HTP

PWR SEQUENCE FPGA:
1/ VCCINT/VCCBRAM: 1.0V
2/ VCCAUX,VCCAUX_IO,VCCO HP BANK: 1.8V
3/ VCCO HR BANK: 3.3V

PWR RATE FPGA:
1/ VCCINT: 1.0V/40A
2/ VCCAUX,VCCAUX_IO,VCCO HP BANK: 1.8V/10A
3/ VCCO HR BANK: 3.3V/10A



PWR SEQUENCE GTX:
1/ VCCINT: 1.0V
2/ VMGTAVCC: 1V
3/ VMGTAVTT: 1.2V & VMGTVCCAUX: 1.8V

PWR RATE GTX:
1/ VMGTAVCC: 1V/8A
2/ MGTVCCAUX: 1.8V SHARE
3/ VMGTAVTT: 1.2V/8A

HTP	project: ***HTPSR5G***		
	Size: A4	Number: 26	Revision: V7
	Date: 26/01/2021	Time: 3:09:13 CH	Sheet 26 of 51
	File: P26 PWR SEQ.SchDoc		

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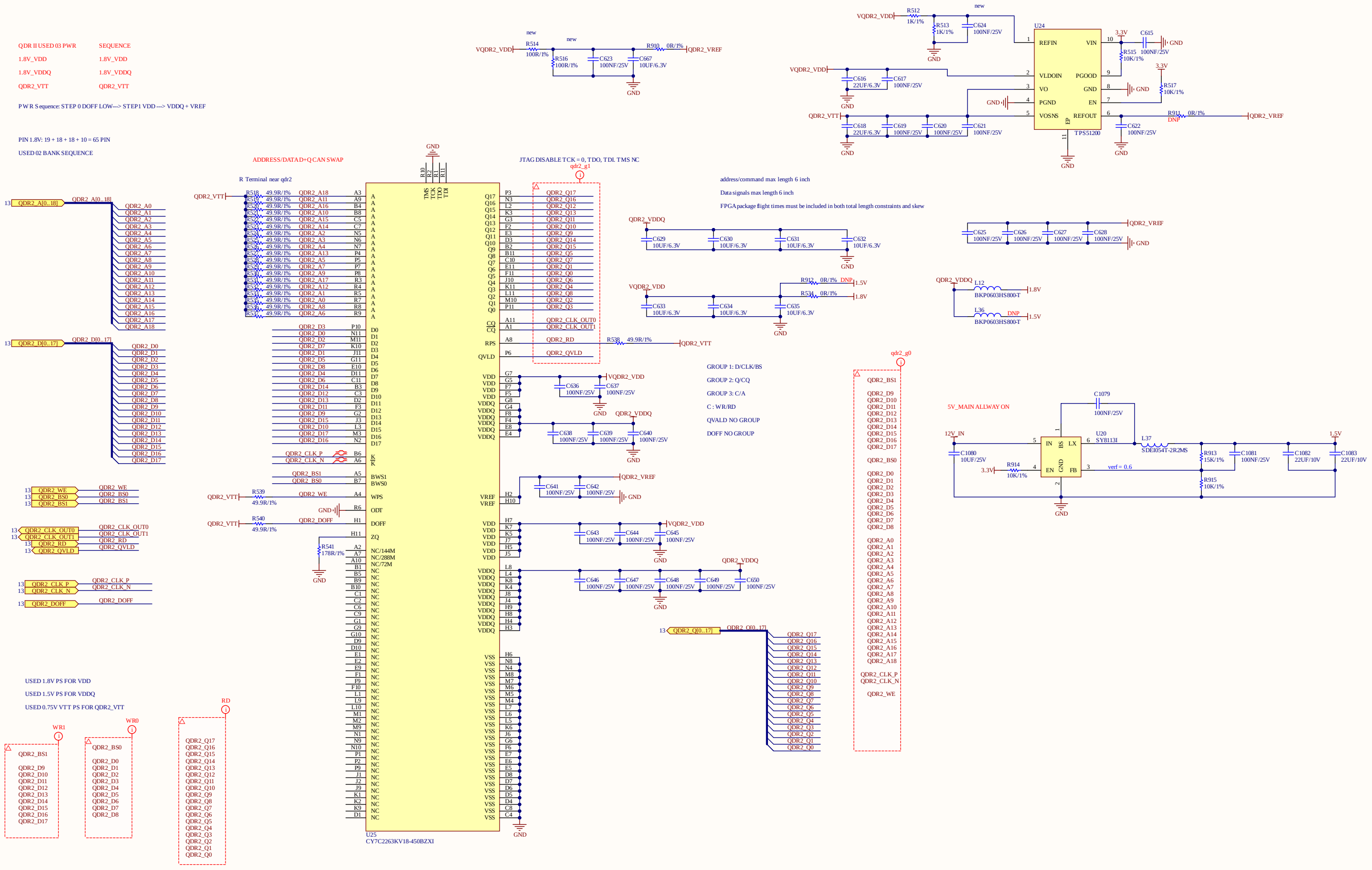
Hanoi



project: ***HTPSR5G***

Size: A4 Number: 27 Revision: V7

Date: 26/01/2021 Time: 3:09:14 CH Sheet 27 of 51

File: P27 QDR II.SchDoc

HTP

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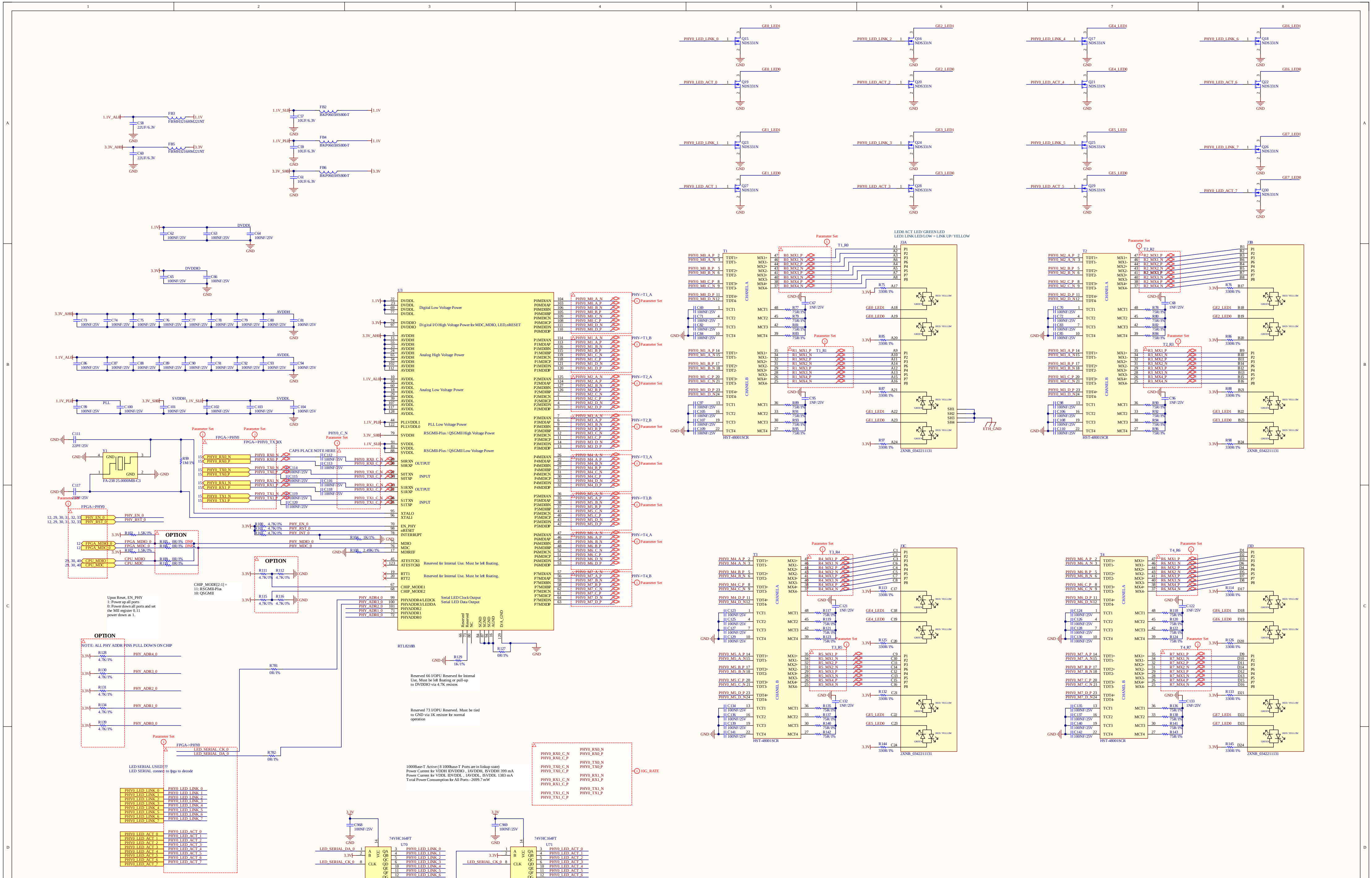

HTP

project: HTPSR5G

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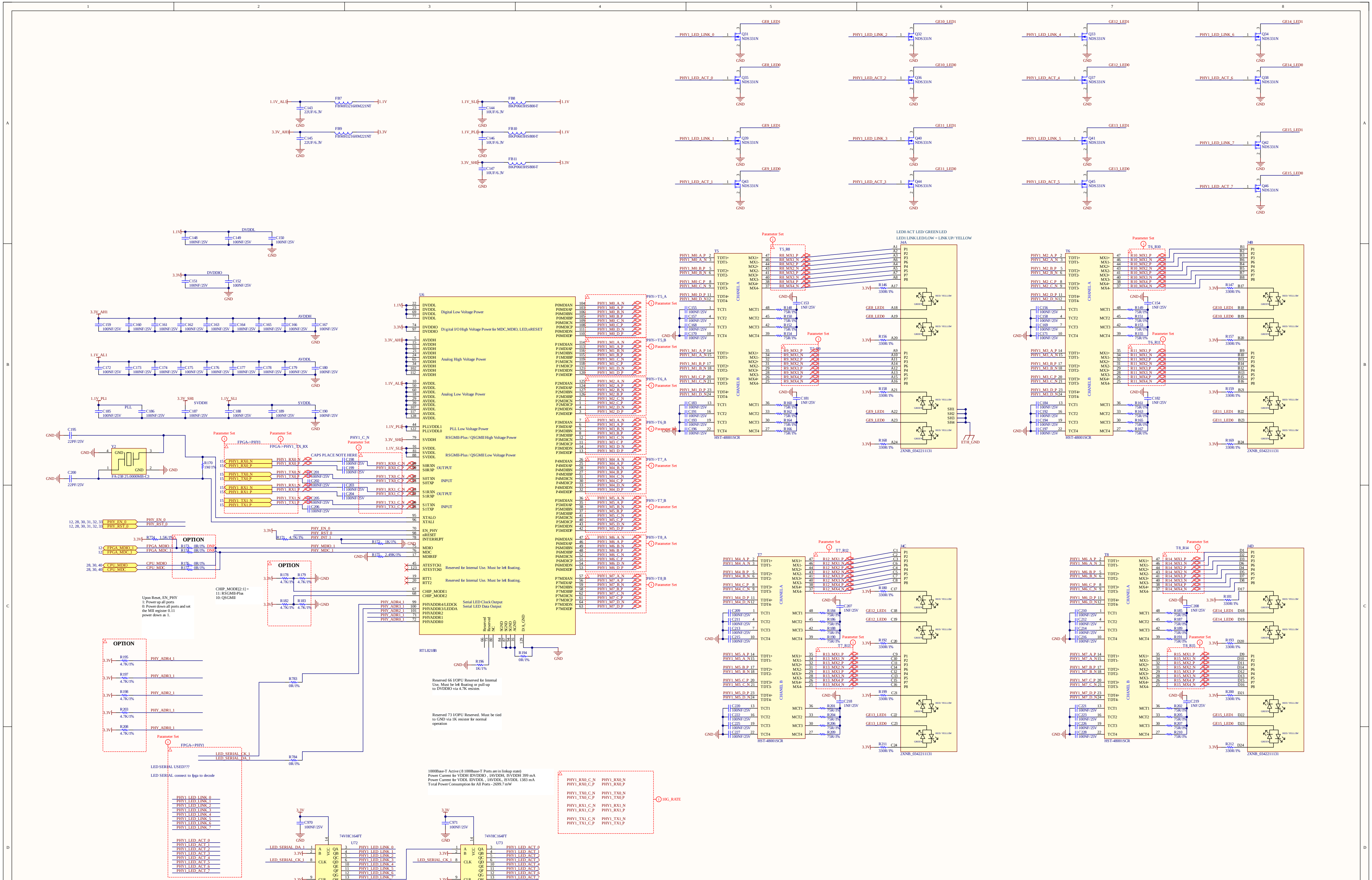
Hanoi

Size: A4	Number: 28	Revision: V7
Date: 26/01/2021	Time: 3:09:14 CH	Sheet 28 of 51
File: P28 PHY0.SchDoc		



project: **HTPSR5G**

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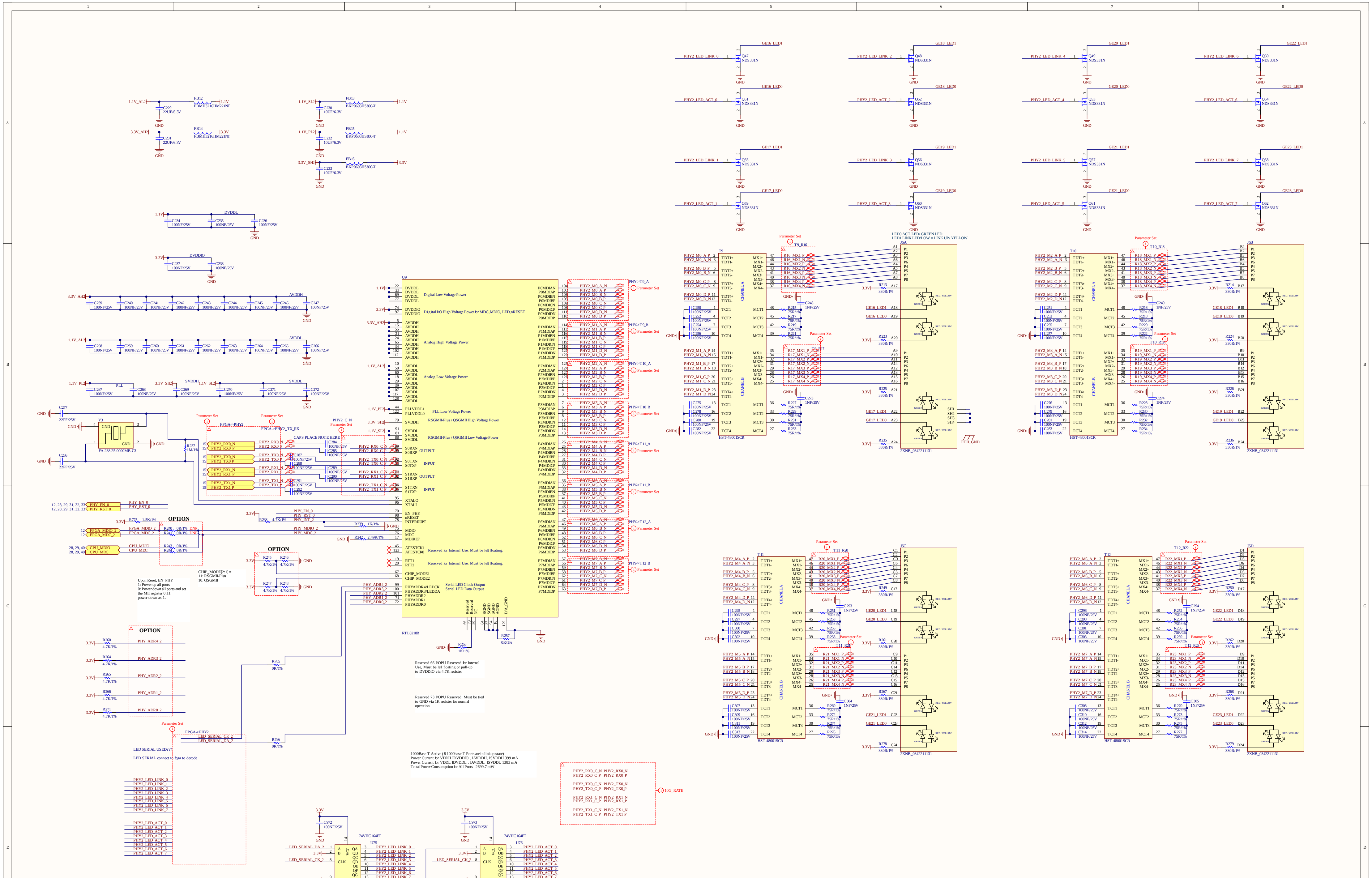
Hanoi

Size: A4	Number: 29	Revision: V7
Date: 26/01/2021	Time: 3:09:16 CH	Sheet 29 of 51
File: P29 PHY1.SchDoc		



project: *HTPSR5G*

Size: A4 | Number: 30 | Revision: V7

Date: 26/01/2021 Time: 3:09:17 CH | Sheet 30 of 51

File: P30 PHY2.SchDoc

HTP

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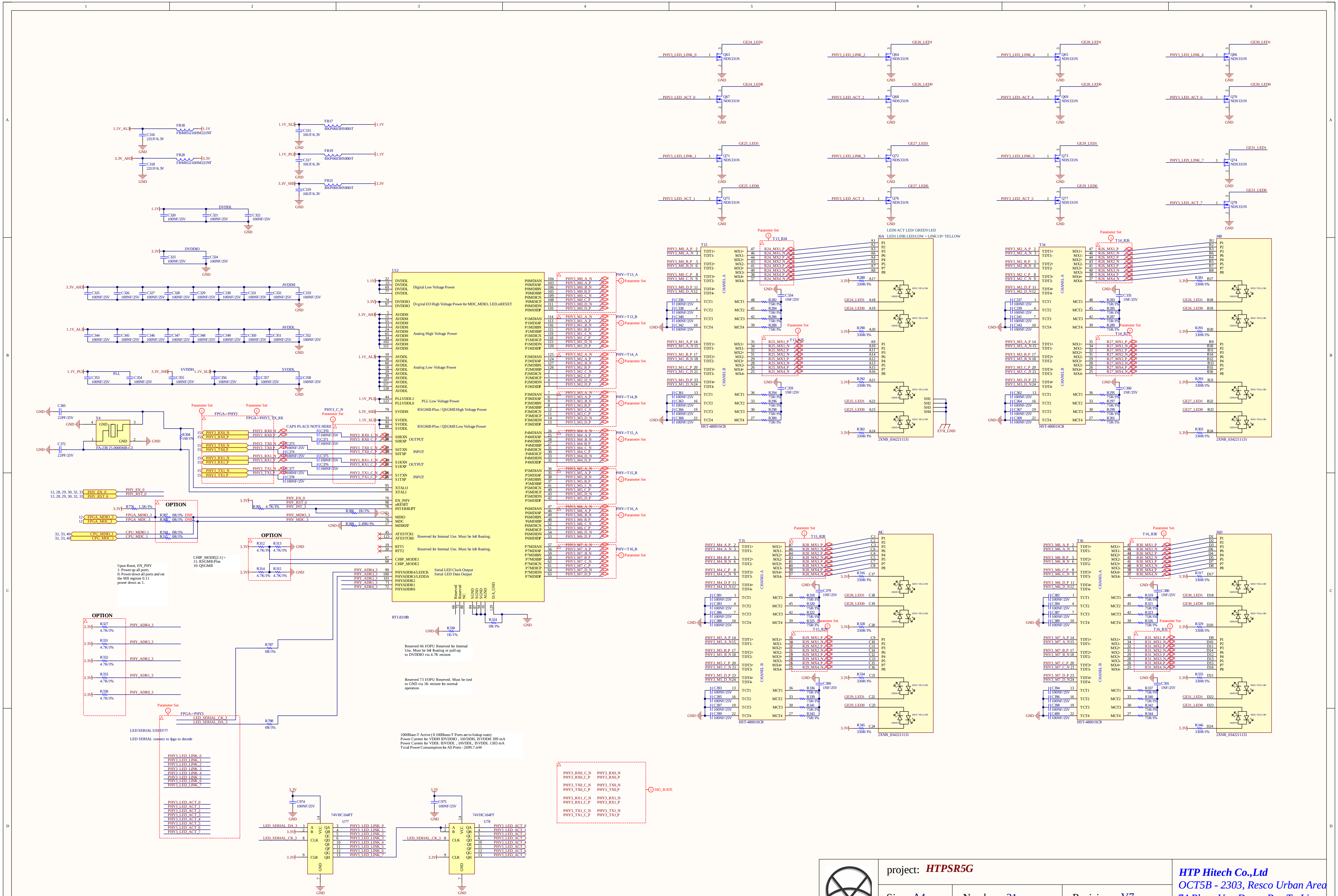
Hanoi



project: HTPSR5G

Size: A4 Number: 31 Revision: V7

Date: 26/01/2021 Time: 3:09:19 CH Sheet 31of 51

File: P31 PHY3.SchDoc

HTP

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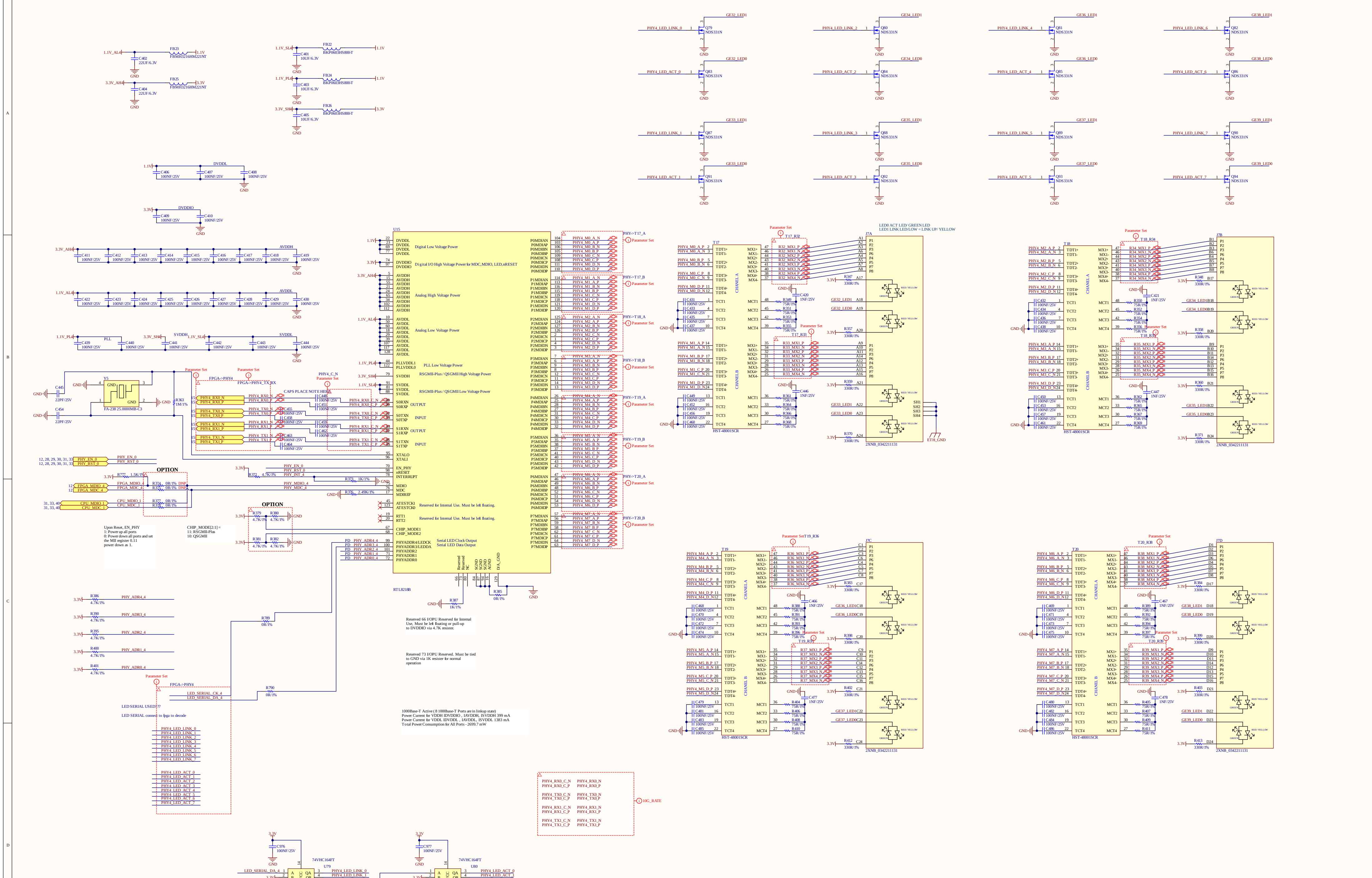
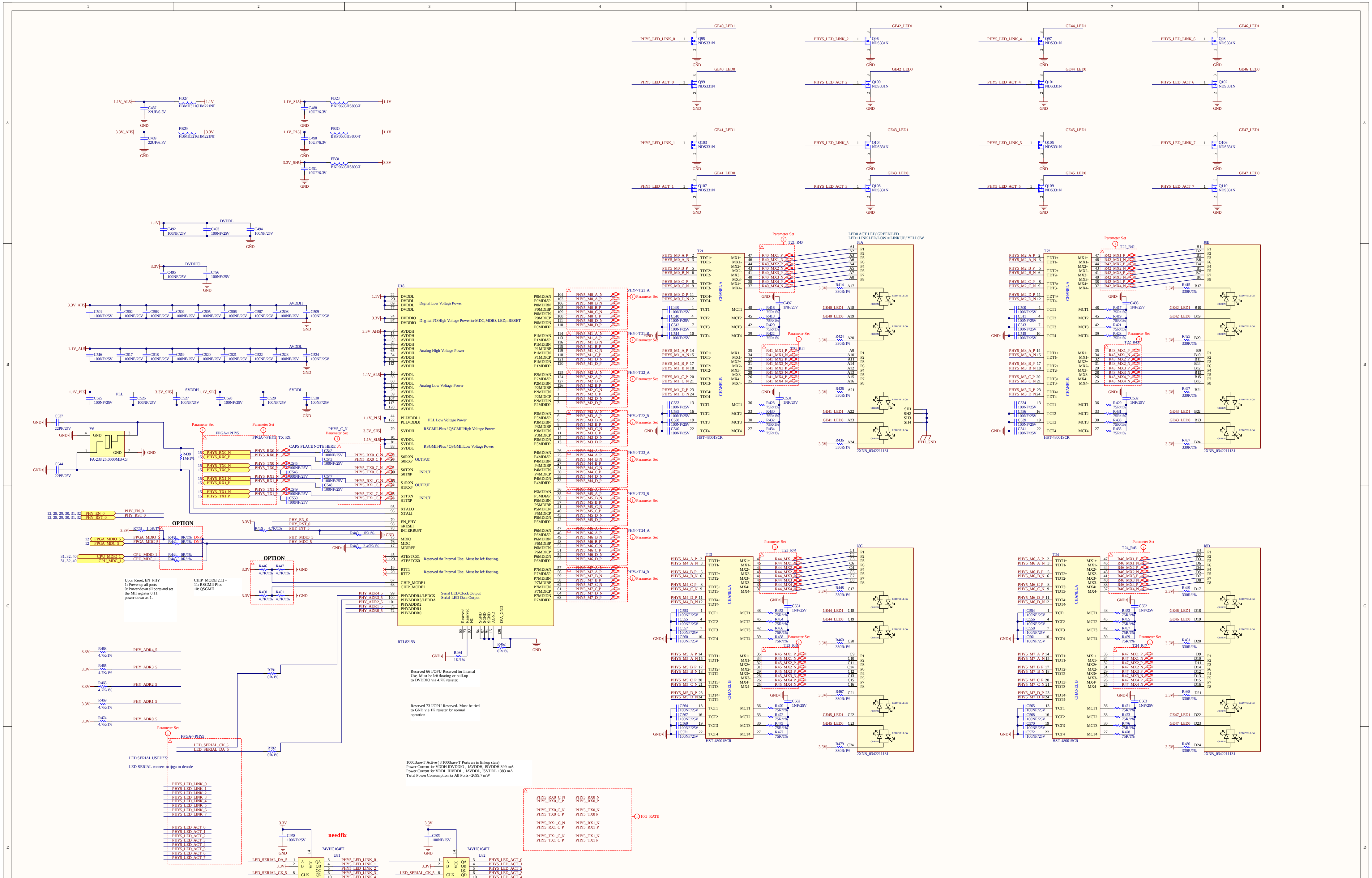
Hanoi



project: **HTPSR5G**	**HTP Hitech Co.,Ltd** *OCT5B - 2303, Resco Urban Area* *74 Pham Van Dong, Bac Tu Liem* *Hanoi*
Size: A4 · Number: 32 · Revision: V7	
Date: 26/01/2021 Time: 3:09:20 CH · Sheet 32of 51	
File: P32 PHY4.SchDoc	

HTP



project:	HTPSR5G
Size: A4	Number: 33
Revision: V7	
Date: 26/01/2021	Time: 3:09:22 CH
Sheet 33 of 51	
File: P33 PHY5.SchDoc	

HTP

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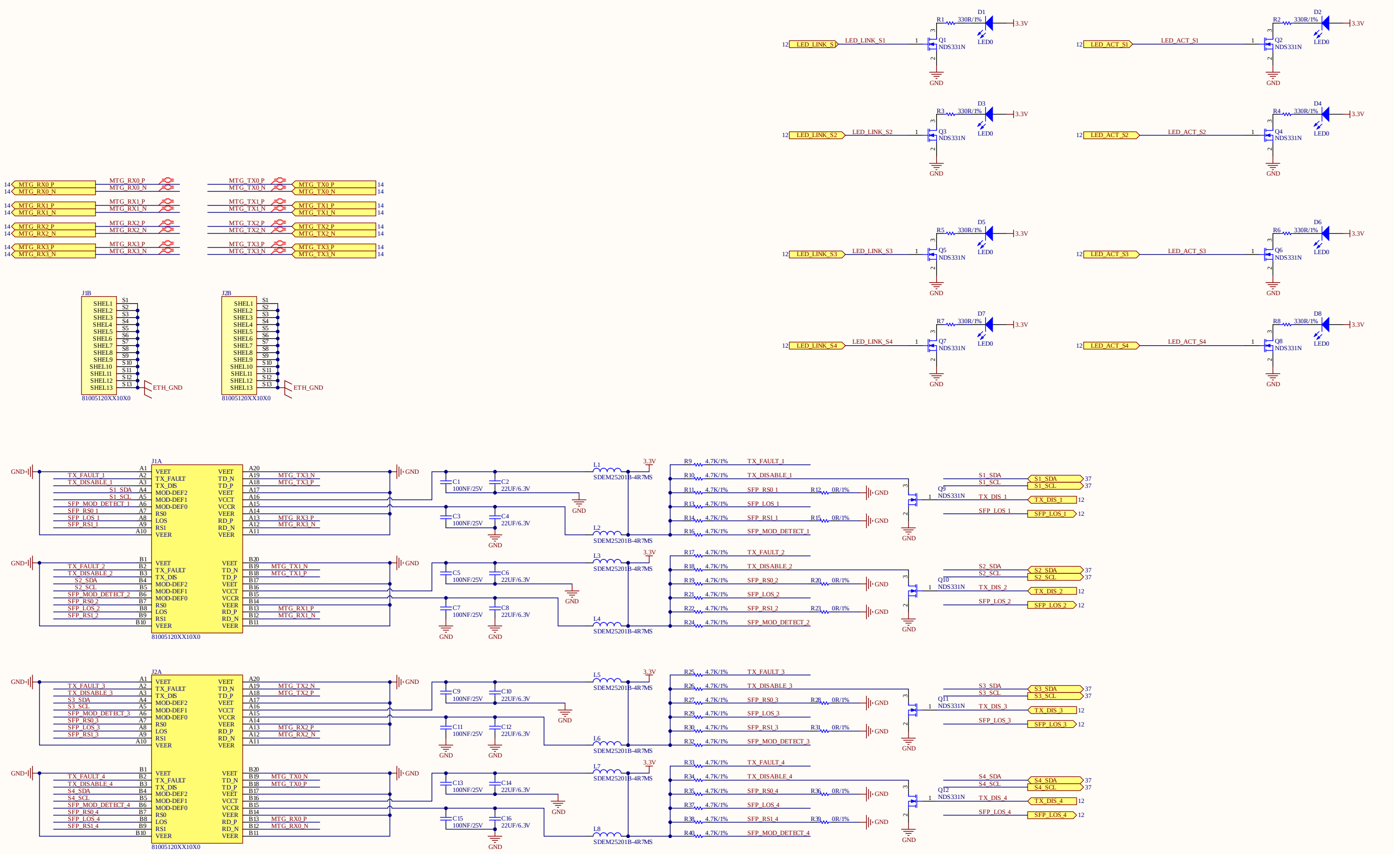
Hanoi



HTP

project: *HTPSR5G*

Size: A4 | Number: 34 | Revision: V7

Date: 26/01/2021 Time: 3:09:23 CH | Sheet 34 of 51

File: P34 4xSFP_SFP+.SchDoc

HTP Hitech Co.,Ltd
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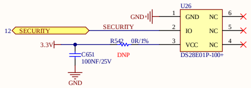
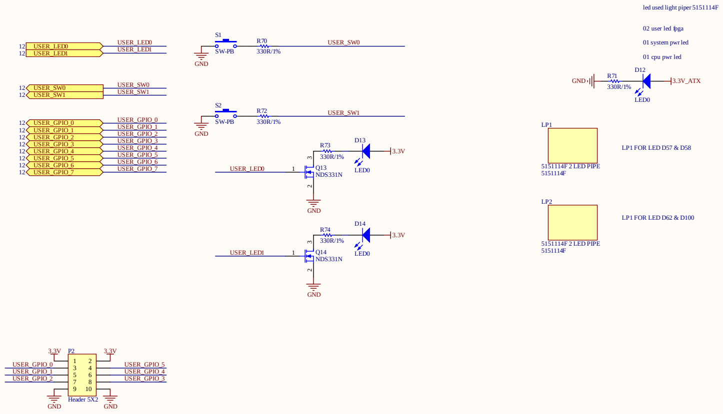
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: *HTPSR5G*			***HTP Hitech Co.,Ltd*** *OCT5B - 2303, Resco Urban Area* *74 Pham Van Dong, Bac Tu Liem* *Hanoi*
Size: A4	Number: 35	Revision: V7	
Date: 26/01/2021	Time: 3:09:23 CH	Sheet 35 of 51	
File: P35 SECURITY CHIP.SchDoc			

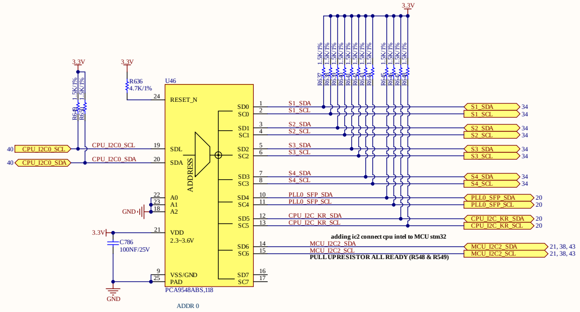
HTP



HTP

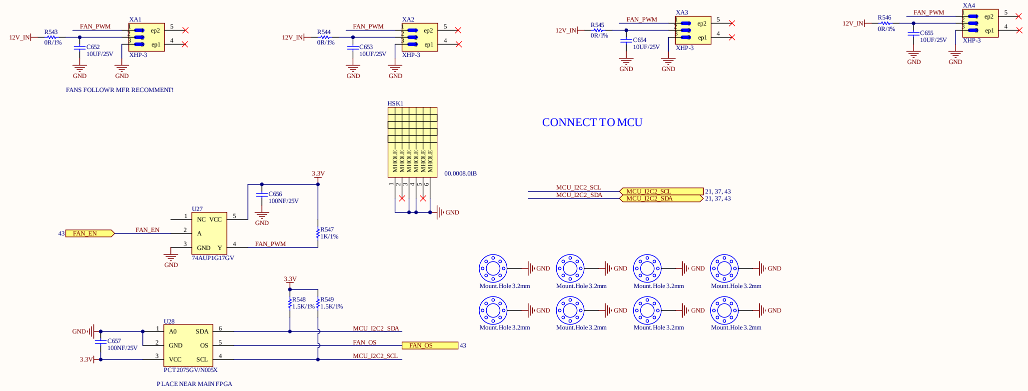
project: ***HTPSR5G***		
Size: A4	Number: 36	Revision: V7
Date: 26/01/2021 Time: 3:09:23 CH		Sheet 36 of 51
File: P36 DEBUG FPGA.SchDoc		

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: **HTPSR5G**			**HTP Hitech Co.,Ltd** OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
Size: A4	Number: 37	Revision: V7	
Date: 26/01/2021	Time: 3:09:23 CH	Sheet 37 of 51	
File: P37 SWITCHS IIC.SchDoc			

HTP



project: HTPSR5G

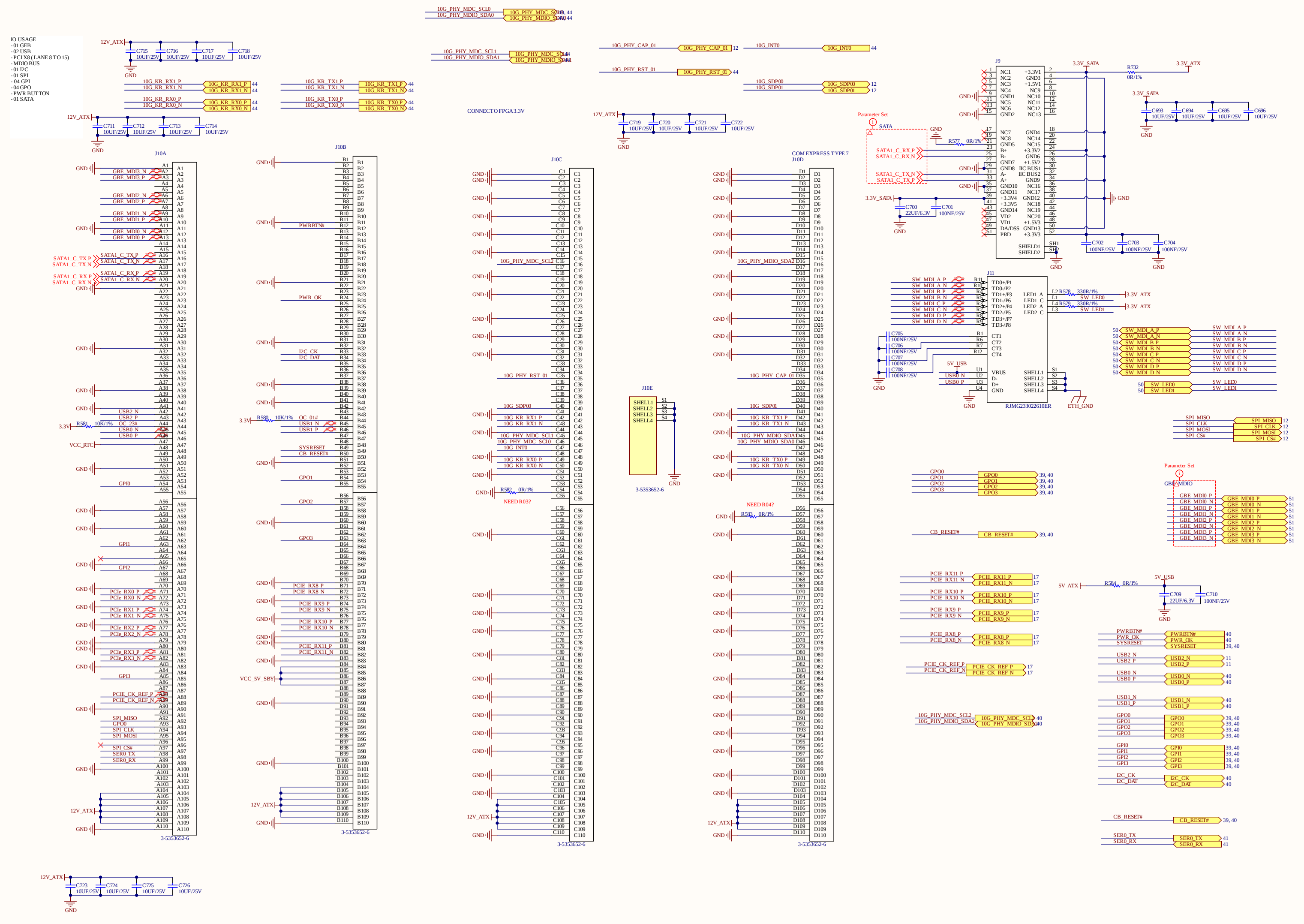
Size: A4 | Number: 38 | Revision: V7

Date: 26/01/2021 Time: 3:09:24 CH | Sheet 38 of 51

File: P38 FANS.SchDoc

HTP

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: *HTPSR5G*

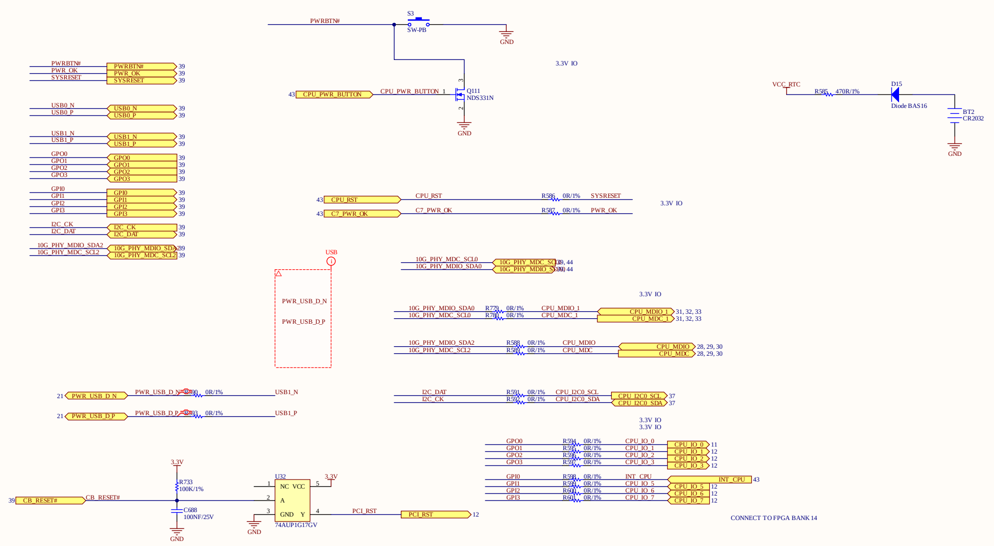
Size: A4 | Number: 39 | Revision: V7

Date: 26/01/2021 Time: 3:09:24 CH | Sheet 39of 51

File: P39 COM EXPRESS TYPE 7 MODULE.SchDoc

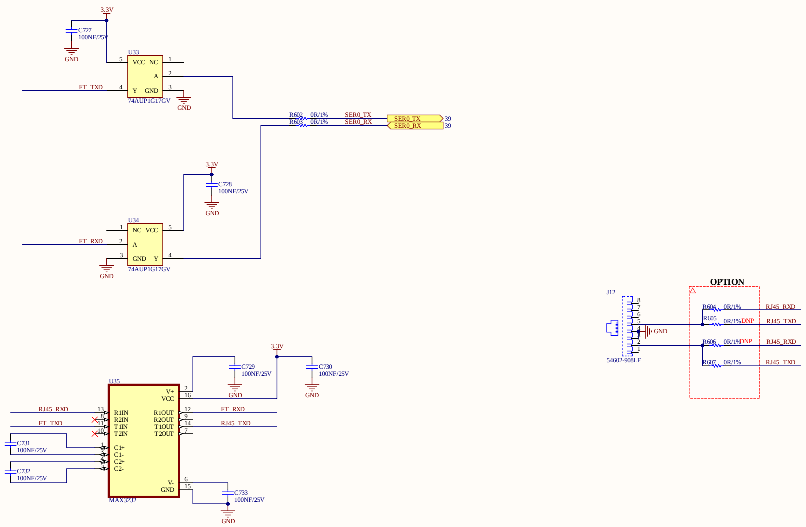
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

HTP



project: **HTPSR5G**			**HTP Hitech Co.,Ltd**
Size: A4	Number: 40	Revision: V7	OCT5B - 2303, Resco Urban Area
Date: 26/01/2021	Time: 3:09:25 CH	Sheet 40 of 51	74 Pham Van Dong, Bac Tu Liem
File: P40 CPU CONN EXT.SchDoc			Hanoi

HTP



HTP

project: HTPSR5G

Size: A4 | Number: 41 | Revision: V7

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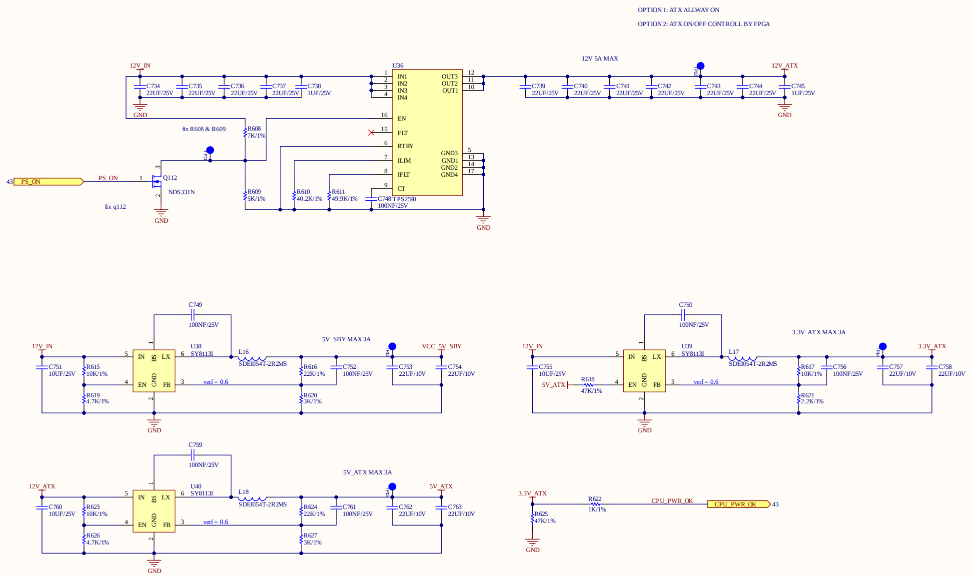
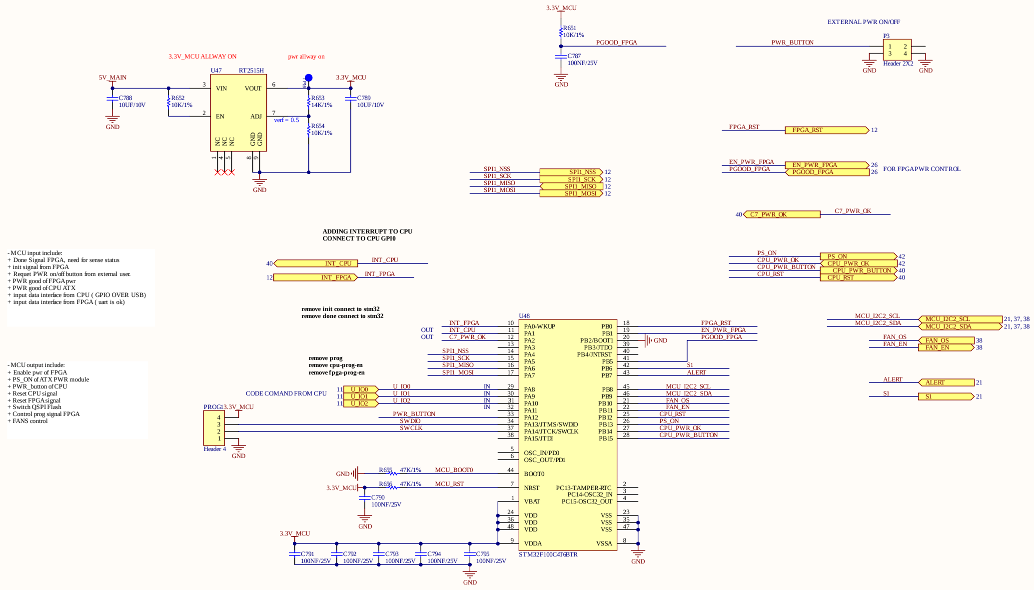
File: P41 CPU CONSOLE.SchDoc

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: *HTPSR5G*			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
Size: A4	Number: 42	Revision: V7	
Date: 26/01/2021	Time: 3:09:25 CH	Sheet 42of 51	
File: P42 CPU PWR.SchDoc			



project: **HTPSR5G**

Size: A4	Number: 43	Revision: V7
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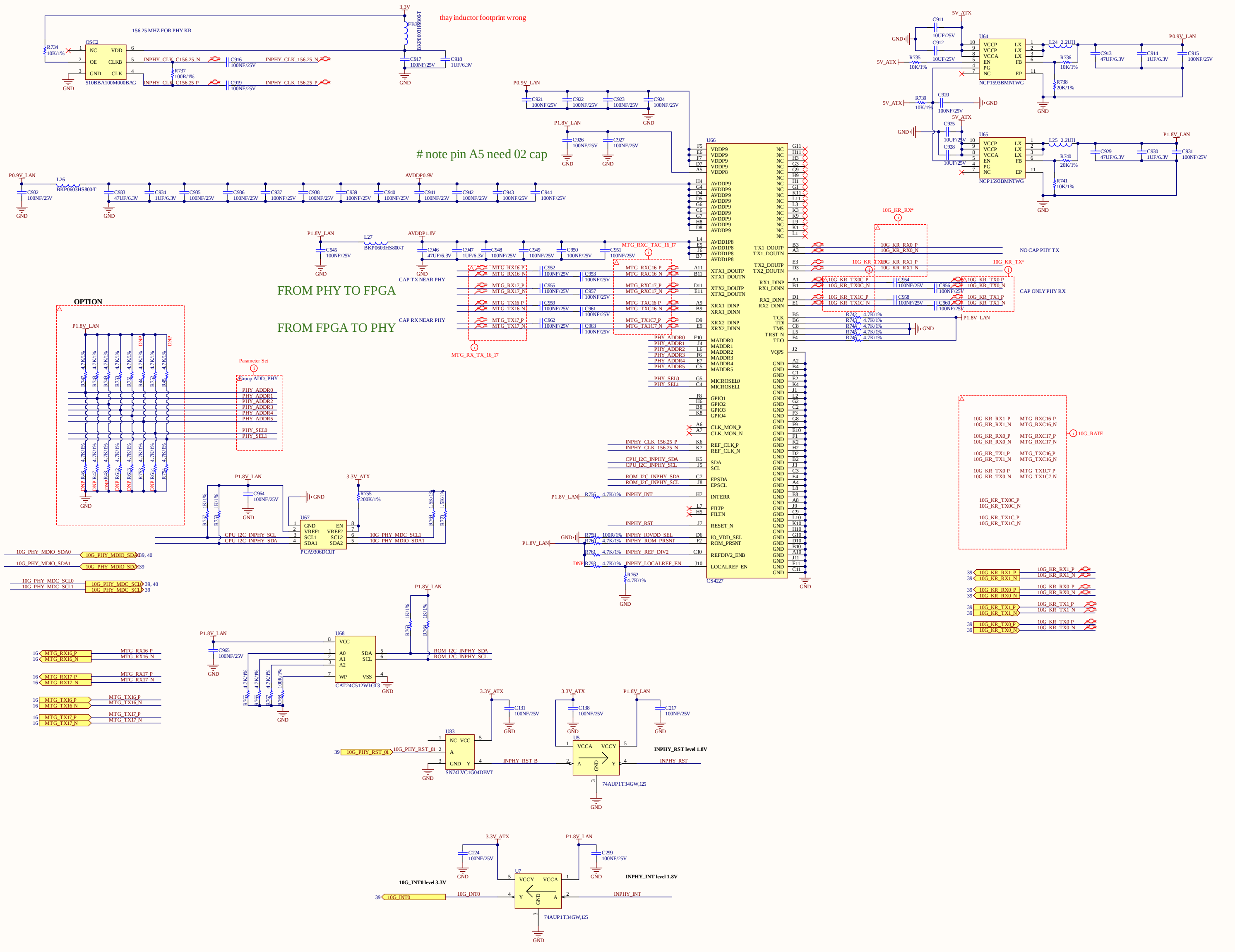
File: P43 SYS CONTROL BLOCK.SchDoc

HTP

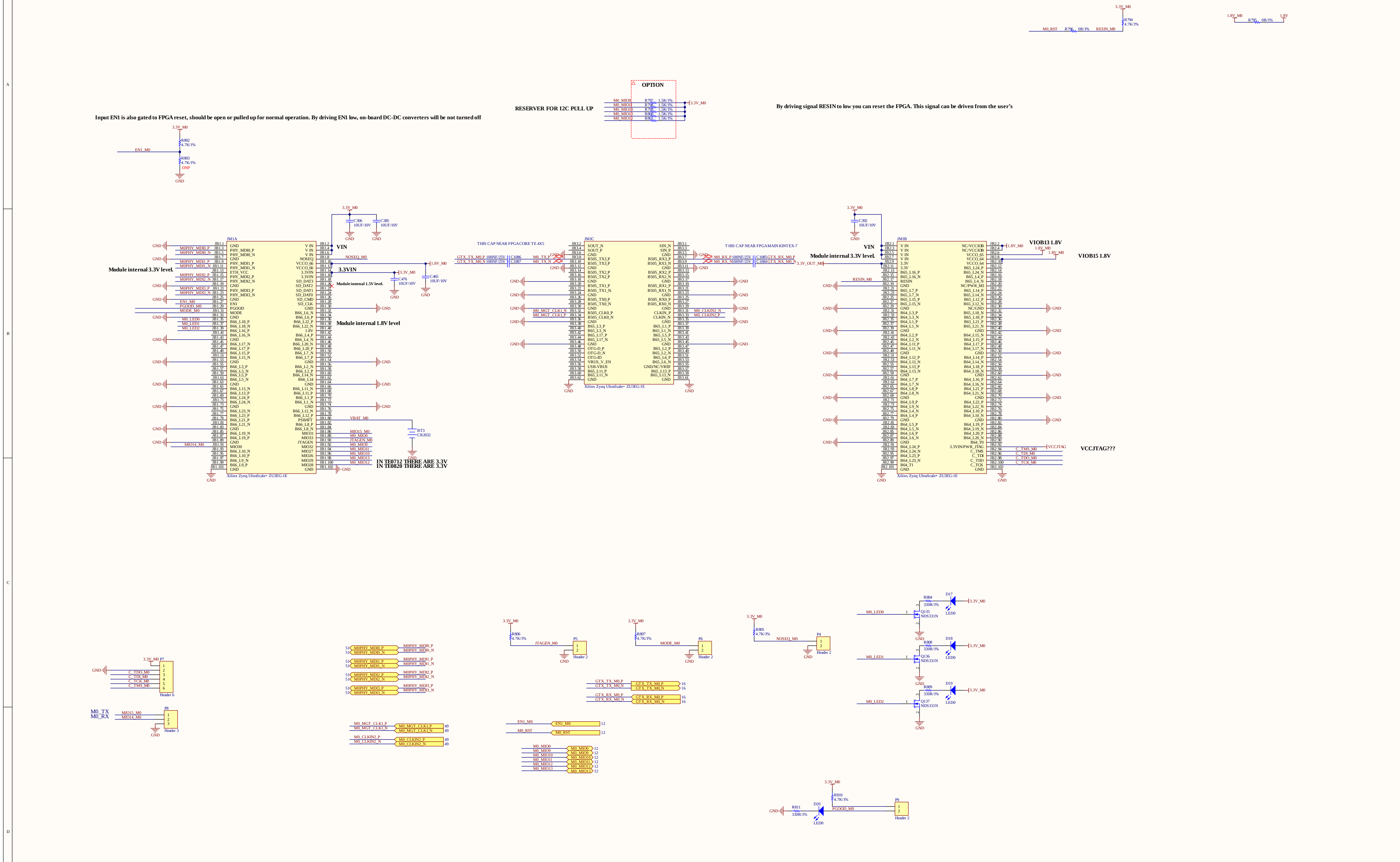
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: **HTPSR5G**

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

Size: A4	Number: 44	Revision: V7
Date: 26/01/2021	Time: 3:09:26 CH	Sheet 44 of 51
File: P44 10G KR PHY.SchDoc		



project: *HTPSR5G*

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

Size: A4 | Number: 45 | Revision: V7

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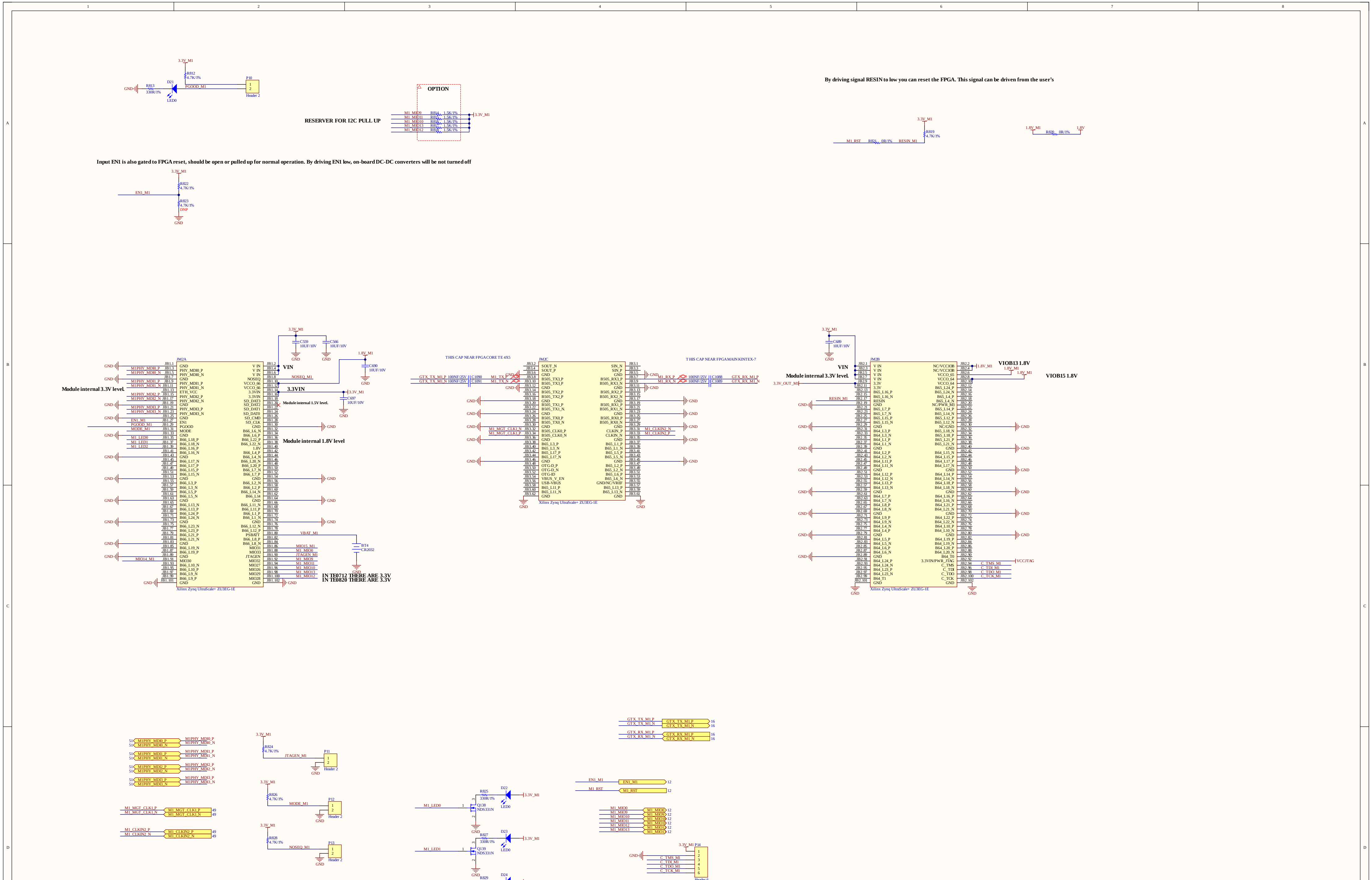
File: P45 4X5 EXT0.SchDoc



project: *HTPSR5G*

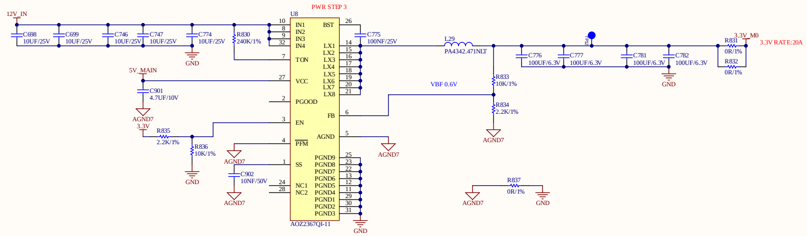
HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi

Size: A4 | Number: 46 | Revision: V7

Date: 26/01/2021 Time: 3:09:27 CH | Sheet 46 of 51

File: P46 4X5 EXT1.SchDoc

HTP



project: ***HTPSR5G***

Size: A4 | Number: 47 | Revision: V7

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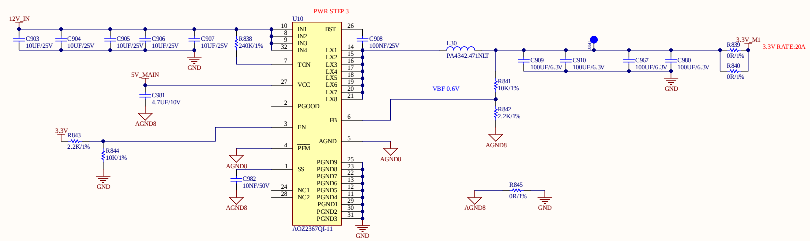
File: P47 PWR M0.SchDoc

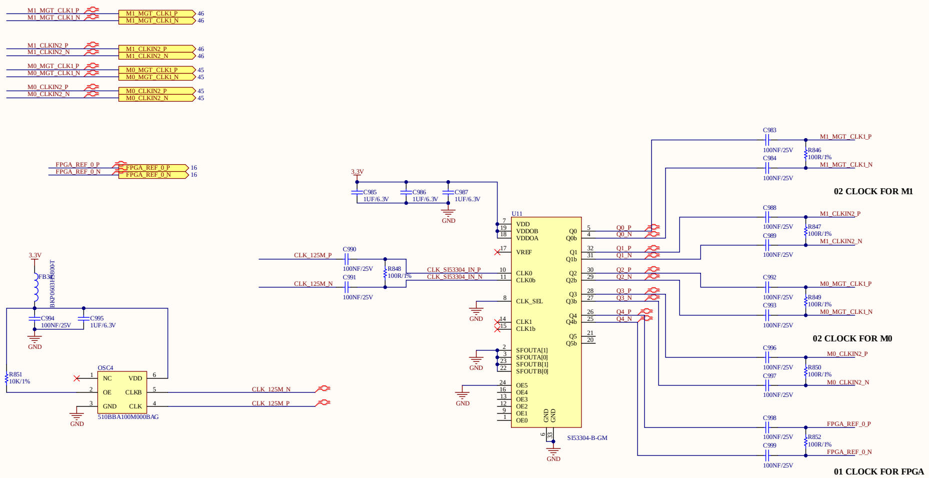
HTP

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: ***HTPSR5G***			***HTP Hitech Co.,Ltd*** *OCT5B - 2303, Resco Urban Area* *74 Pham Van Dong, Bac Tu Liem* *Hanoi*
Size: A4	Number: 48	Revision: V7	
Date: 26/01/2021	Time: 3:09:28 CH	Sheet 48 of 51	
File: P48 PWR M1.SchDoc			

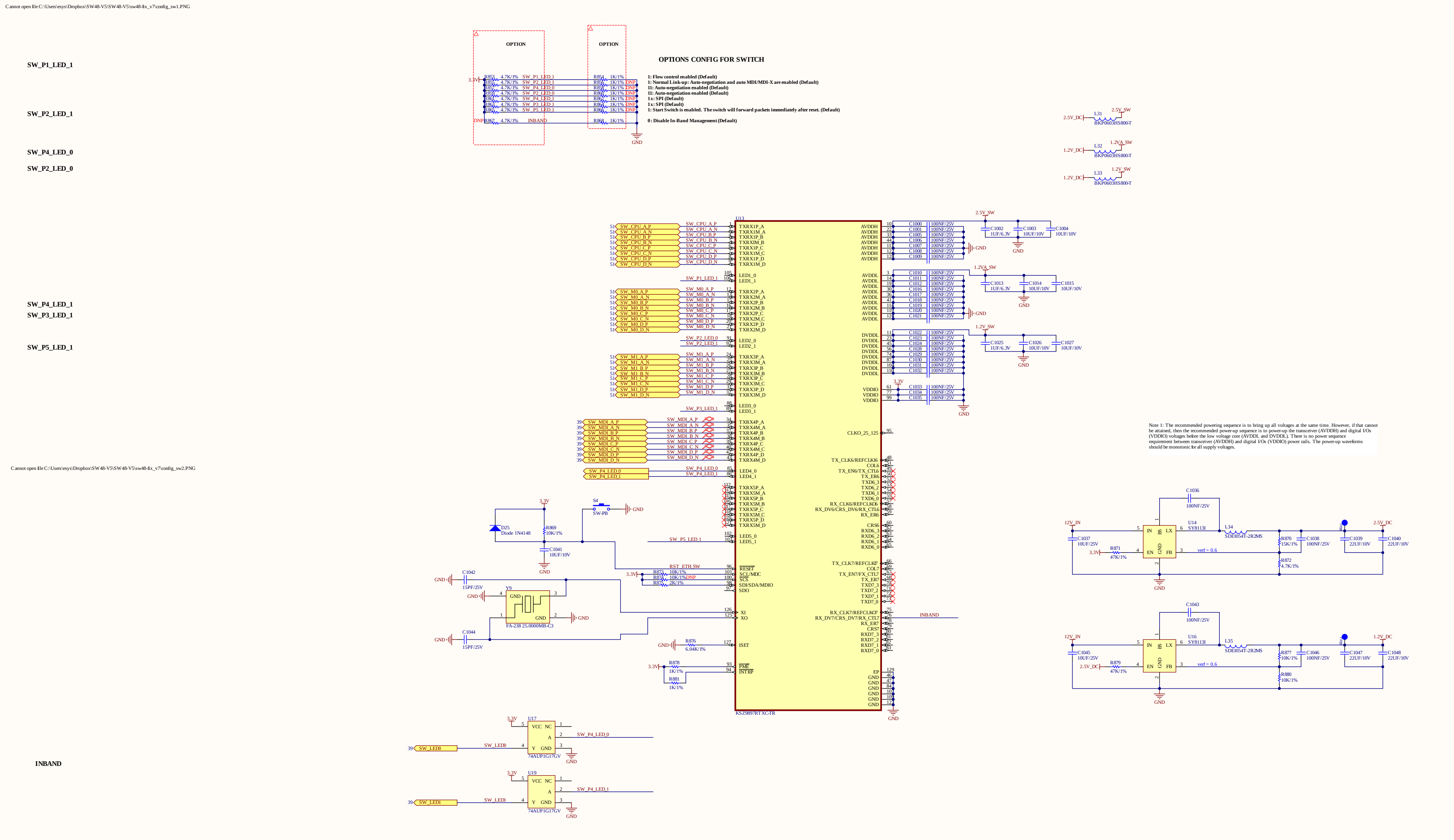
HTP



The output enable pin has an internal pull-up that enables the outputs when left unconnected

project: HTPSR5G			HTP Hitech Co.,Ltd
Size: A4	Number: 49	Revision: V7	OCT5B - 2303, Resco Urban Area
Date: 26/01/2021	Time: 3:09:28 CH	Sheet 49 of 51	74 Pham Van Dong, Bac Tu Liem
File: P49 CLOCK & BUF.SchDoc			Hanoi

HTP

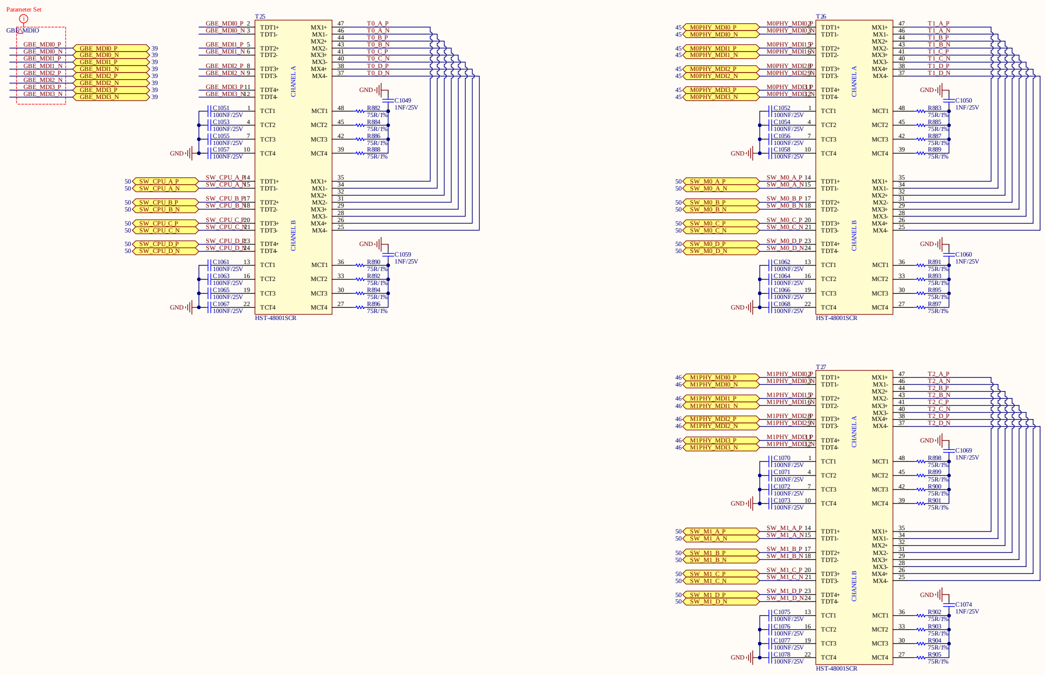


project: *HTPSR5G*

Size: A4	Number: 50	Revision: V7
Date: 26/01/2021	Time: 3:09:28 CH	Sheet 50 of 51
File: P50 SW ETH.SchDoc		

HTP

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi



project: **HTPSR5G**

Size: A4 | Number: 51 | Revision: V7

Date: 26/01/2021 Time: 3:09:29 CH | Sheet 51 of 51

File: P51 TRANS.SchDoc

HTP

HTP Hitech Co.,Ltd
OCT5B - 2303, Resco Urban Area
74 Pham Van Dong, Bac Tu Liem
Hanoi